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Deep learning-based de novo discovery and design of therapeutics that reverse disease-associated transcriptional phenotypes

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Author contributions

J.X. conceptualized the study, developed software, performed data analysis, and led the HCC study. D.L. performed data analysis and led the IPF study. M.T. performed experiments in the HCC study under the supervision of S.S. and M.C.. M.S. proposed and developed the algorithms including RCL, GNN-based RCL and MolSearch under the supervision of J.Z. and B.C.. R.C. evaluated the RCL algorithms. M.A., L.H., and K.U. performed experiments in the IPF study under the supervision of X.L.. S.P., D.L., and R.S. developed the web portal and provided bioinformatics support. E.L., B.A., M.G., R.N., and E.E. contributed to assay development, drug screening, compound synthesis, and PK studies. R.G., T.J., and C.L. provided biospecimens for the IPF study. J.X., D.L., M.T., M.S., X.L., M.C., J.Z., and B.C. drafted the manuscript with input from others. B.C. conceptualized and supervised the study.

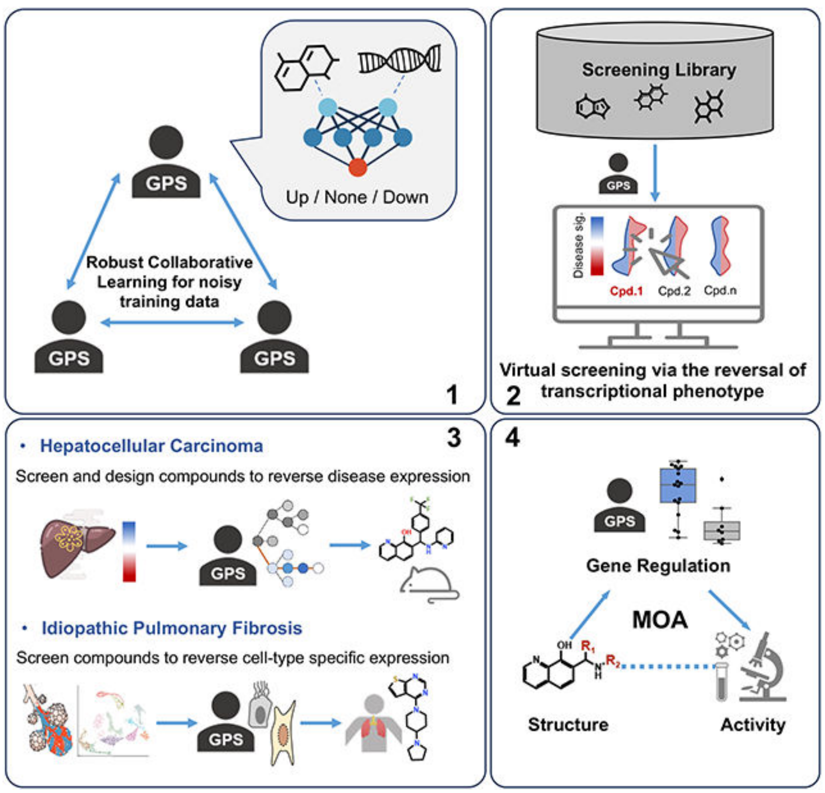
Declaration of interests

An invention disclosure has been filed (B.C., X. L., D.L., and R.G.).

Summary

Identifying drugs that reverse disease-associated transcriptomic features has been widely explored for drug repurposing, but its potential for de novo drug discovery remains underexplored. Here, we present GPS, a deep learning-based drug discovery platform, guided by transcriptomic features, that screens large compound libraries and optimizes lead molecules. We first develop a model that captures transcriptomic perturbation signatures solely from chemical structures and deploy it to library compounds. We refine scoring methods and employ a tree search method for optimization. By incorporating Structure-Gene-Activity Relationships, we uncover drug mechanisms from transcriptomic data. We evaluate GPS across multiple diseases and conduct extensive validation in two cases. In hepatocellular carcinoma, we discover two unique compound series with favorable cellular selectivity and in vivo efficacy. In idiopathic pulmonary fibrosis, we identify one repurposing candidate and one novel anti-fibrotic compound by reversing gene expression of multiple distinct cell types derived from single cell transcriptomics.

Graphical Abstract



Keywords

Transcriptome reversal; virtual screening; de novo drug design; artificial intelligence; curriculum learning; structure-gene-activity relationship; polypharmacology; hepatocellular carcinoma; idiopathic pulmonary fibrosis

Introduction

Current virtual drug screening studies are predominately based on docking against specific protein targets or artificial intelligence (AI) and machine learning models trained on screening data[1–5]; few studies utilize rich transcriptomic features routinely used to characterize diseases and cellular states, made possible by advances in transcriptomics technologies, particularly single-cell RNA-seq[6]. Identifying drugs that reverse expression of disease-associated transcriptomic features has been widely explored as a strategy for discovering drug repurposing candidates[7–11]. A drug candidate should decrease the expression of up-regulated disease genes and increase the expression of down-regulated disease genes, thereby restoring a healthy transcriptional phenotype. However, this approach is limited to compounds that are already profiled in databases[7,12], and does not support novel compound screening and optimization, thereby limiting its widespread adoption in early drug discovery.

To implement the above gene expression reversal approach for screening ultra-large compound libraries like those commonly used in target-based screening[13], gene expression profiles of the library compounds are required. However, producing such profiles for a vast number of compounds under various biological conditions is currently not feasible. Nevertheless, the massive existing drug-induced gene expression profiles have enabled the development of advanced machine learning models to infer gene expression based solely on chemical structures. Recent efforts have demonstrated the feasibility of predicting compound-induced gene-expression profiles[14–19] and the potential of using this approach in preclinical drug discovery[20]. However, these studies only included small screening libraries of commonly studied compounds, and they did not explore novel compounds or perform lead optimization, an essential step in early drug discovery.

Here, we present a deep learning-based drug discovery system for the screening of large compound libraries and *de novo* design of compounds that can reverse transcriptional phenotype (Figure 1A). We first trained a Gene expression profile Predictor on chemical Structures (GPS) and then used the model to predict transcriptomic perturbation signatures compising gene regulation (up/down/no effect) across the transcriptome for millions of compounds in the ZINC library and the Enamine HTS library. To improve the prediction, we proposed a collaborative learning model to correct input gene expression data. We next enhanced the compound scoring function and developed a simple, yet effective tree-search method for multi-objective optimization. Finally, we proposed Structure-Gene-Activity Relationship (SGAR) analysis, for the elucidation of drug mechanism from the rich transcriptomic data. To demonstrate the utility of the system, we identified and validated compounds for hepatocellular carcinoma (HCC), and idiopathic pulmonary fibrosis (IPF). HCC is the sixth most common cancer and the third leading cause of cancer-related death worldwide[21]; it has a high mortality rate, yet lacks effective treatment options. IPF is a rare chronic lung disease characterized by fibrosis of the lung; it currently has no curative treatment options and the median survival for IPF patients after diagnosis is approximately three years[22]. Thus, new effective therapeutics are urgently needed for both diseases. In HCC, we designed a highly selective and potent compound, while in IPF, we discovered a repurposing candidate and screened novel compounds targeting multiple cell populations that are likely associated with disease progression.

Results

Prediction of compound-induced transcriptomic signatures based on chemical structures

The core component of transcriptomics-based drug design is to predict compound-induced gene expression signatures based on their chemical structures. To develop a high-performance model, it is essential to address the technical and biological variations inherent in high-throughput profiles.

For instance, the average correlations between LINCS Phase I Level 4 replicates are only 0.5 (Figure S1A and S1B), and the median reproducibility score is less than 0.6 between Phase I and Phase II profiles (Figure S1C). This issue was addressed during both pre-processing and training steps. During pre-processing, drug-gene interactions were categorized into three groups: down-regulation (z-score < -1.5), up-regulation (> 1.5), and no-effect, to reduce the noise in expression values. Then, the featured genes whose expression can be predicted from chemical structures were identified by a random forest model (STAR Methods). In total, 307 out of the 978 landmark genes were found to be predictable, with more than half shared across different cell models (Figure S1D and Data S1). GPS was trained using the Phase I LINCS data, including 18,746 compound effects on the 978 landmark genes of four commonly studied cell lines HEPG2, MCF7, PC3 and VCAP. During training, we adopted a curriculum learning framework called Robust Collaborative Learning (RCL) to re-weight high-quality data points (i.e., drug-gene pairs) rather than discarding the massive low-quality drug transcriptomic profiles, through knowledge fusion from multiple experts (Figure 1A).

In GPS, a neural network takes the structural fingerprint of a compound and the Gene Ontology (GO) terms of a gene as the input, and learns whether this compound could up-regulate, down-regulate, or have no effect on the expression of the input gene. The learned knowledge was shared among several peer networks for iterative noise control on the labels, and eventually utilized approximately 80% of all data points for training. The proposed RCL framework achieved significant improvement as compared to baseline methods in both internal (random-split) and external validation (new compounds) (Figure 1B and Table S1). Incorporating gene information and modeling all genes together instead of treating each gene as an individual task (RF-SG-HQ vs RF-HQ) can largely reduce the overfitting problem in single gene models. Simply modeling all genes together using a vanilla multi-task learning model did not substantially mitigate this issue (RF-SG-HQ vs MTL). Training with only statistically high-quality profiles (RF-HQ) or using all available drug profiles (DeepCop) resulted in suboptimal generalization performances compared to the proposed RCL method. The overall performance differences between RCL and baseline methods are all significant ($P < 1E\text{-}08$) across the four cell models. Compound-wise accuracy distribution was significantly better aligned with experimental replication as compared to a random model (Figure S1C). The performance is independent of chemical similarity between training and testing compounds (Figure S1E). However, the performance was suboptimal under experimental conditions other than 10μM concentration and 24h treatment, likely due to the smaller sample size (Figure S1F). Regarding the training algorithms: i) RCL outperformed major ML/DL methods (Figure 1B); ii) it also outperformed robustness-based advanced deep learning methods regardless of compound embedding methods (Table S2); iii) chemical embeddings learned by a graph neural network[23] and transformer-based models did not outperform structural fingerprints (Tables S2, S3, and S4), and random forest models performed the best for “seen” compounds but the worst for “unseen” compounds which were not the in the training set (Table S3).

Biological relevance of GPS-imputed compound-induced transcriptomic signatures

While training data included expression changes of 978 landmark genes, the whole transcriptome contains over 20,000 protein-encoding genes. To expand the predictive coverage, we embedded genes using 1,107 GO features, enabling the prediction of the whole transcriptomic signatures based on GO features. To identify genes with confident predictive accuracy, we utilized a calibration dataset (SRA: DRP006006) containing whole transcriptome profiles for 45 drugs across 23 cell lines. Predicted expression changes were compared with true changes, leading to the identification of 2,018 well-predicted genes with balanced accuracy values greater than 0.5 and macro F1 scores greater than 0.4 (Data S2). Finally, combining these with the pre-selected training genes, our pipeline could predict compound-induced transcriptomic signatures of 2,198 genes with high confidence (Figure 2A). GO enrichment analysis (Figure 2B) of these genes suggested that more than 26% of them were located in the nucleoplasm (fold enrichment = 1.52, FDR = 4.72E-27). The enriched biological processes were related to cell cycle regulation, which corroborated with the significant enrichment of the cellular component - spindle (fold enrichment = 2.05, FDR = 1.42E-04). These results imply that gene expression related to cell cycle, transcription, and kinase signaling is relatively more responsive to compound perturbations. The differing performances between selected and non-selected genes suggested their distinct transcriptional regulation. A comparison of the basal expression level of selected and non-selected genes revealed that the former generally exhibited higher expression levels, not only in the four GPS cell models but also in other cell lines (Figure 2C). In general, although profiled under different cellular contexts, the transcriptomic signature induced by the same compound tend to be conserved (Figure 2D, Data S3), and correlated across different cell lines (Figure 2E, Data S4). Transcriptional profiles of the same compound across cellular contexts are aggregated into a robust profile, which is highly correlated with the profiles from individual cell lines (Figure 2F, Data S4). The aggregated profile from the four cell models covered by GPS is also correlated with a profile aggregated from other cell lines (Figure 2F, Data S4).

To evaluate whether the profiles generated by GPS reflect the biological processes affected by a compound treatment, we interrogated the GPS-predicted transcriptional profiles (aggregated over the four cell models) of the small molecule inhibitors against nine therapeutic targets across three well-studied pathways: Wnt, Ras-Raf, and NF-κB (Data S5), as well as their structural fingerprints. When visualizing their similarities in chemical space, compounds sharing the same target clustered together (Figure 2G) because of the conservative pharmacophore of the protein binding pocket. However, in transcriptional space, a different pattern emerged. Rather than forming target-specific clusters, compounds modulating the same pathway grouped together (Figure 2H). In fact, compared with chemical features, transcriptional features could recall more neighboring compounds inhibiting different targets in the same pathway (Figure 2I), suggesting that the predicted expression profiles captured drug-related biological processes. We also investigated 20 protein targets from various classes, e.g., a transporter ABCC1, an epigenetic eraser HDAC1, and an ion channel SCN9A, together with dozens of inhibitors for each target. Based on the similarities between GPS-predicted compound profiles and gene knockdown profiles in LINCS (aggregated over different experiments, Data S6, similarities scored with rank sums p-values), inhibitors for eight out of 20 targets showed significantly higher similarity scores to their respective knockdown profiles than non-inhibitors (Figure 2J), indicating the potential of using GPS-predicted gene expression profiles to identify the direct targets of compounds through simple statistical methods.

To conduct large-scale transcriptomics-based compound screening, we prepared a virtual library of compound-induced gene expression signatures covering seven million drug-like compounds in the ZINC database, a catalog of compounds available from multiple vendors. While there is some overlap in the drug-like chemical space between this library and the training compounds, a substantial portion of the library explores new chemical space (Figure 2K).

Validation of gene expression reversal using GPS-predicted compound profiles

After the in silico modeling, we next aimed to use the GPS platform to discover novel and selective anti-HCC compounds. To achieve this, a previously defined HCC signature[8,11] was queried against the GPS-generated transcriptomic profiles for compounds in the ZINC library with almost seven million drug-like compounds (Figure 3A). Then the top-ranked compounds were nominated for testing their cytotoxicity on HCC cell lines and primary hepatocytes *in vitro*, and their efficacy *in vivo*.

Before initiating the expensive screening campaign, we evaluated whether the predicted compound profiles and the reversal scoring function could be adopted for screening in HCC and other diseases by leveraging published data. The compound-HCC signature reversal was first calculated using RGES, which was adapted from the Connectivity Score[7,10,12] and has been proven effective in repurposing profiled drugs for various diseases[8,11,24–26]. However, raw RGES values are sensitive to the number of genes affected by a compound, making it unsuitable for comparing compounds with varying gene set sizes predicted by GPS. Thus, we developed Z-RGES which normalizes RGES by sampling the distribution of raw RGES values for the same number of up- or down-regulated genes and then applying Z-transformation of the raw RGES (Figure 3A). A more negative Z-RGES indicates a higher ability to reverse the disease signature, thus suggesting better efficacy. Z-RGES demonstrated a significant correlation with anti-HCC cellular activity (Spearman $R = -0.554$, $P = 0.0049$, Figure 3B), whereas raw RGES showed no significant correlation (Figure S2A). Z-RGES also outperformed other published approaches (Figure S2, STAR Methods). In the CTRP HepG2 drug sensitivity dataset, Z-RGES achieved an AU-ROC of 0.768 (Figure 3C) and a top hit rate of 40% (Figure 3D), whereas the raw RGES performed randomly. Compared with the sRGES in OCTAD[27], derived from high quality drug profiles in the original LINCS database, Z-RGES showed a slightly higher top 10 hit rate in the

PRISM[28] Huh7 drug sensitivity data set (Figure 3E), suggesting a higher successful rate in large-scale virtual screening campaigns.

In addition to HCC, we previously observed that the reversal of cancer gene expression correlates with drug efficacy in colon adenocarcinoma (COAD)[8]. Because these drugs were profiled in LINCS, we expanded the analysis by using GPS-imputed drug profiles. Using the COAD disease signature (log2 fold change > 1 and adjusted $P < 0.05$), we computed Z-RGES of these drugs. We also observed that active compounds have lower Z-RGES than inactive ones (Figure S2C). Interestingly, while using the entire COAD disease signature (with any cutoff), we did not observe a significant difference, highlighting the importance of the disease signature, which was extensively explored elsewhere[27].

We further evaluated this approach in Alzheimer's disease (AD), a widely studied disease with extensive preclinical and clinical drug testing data. We first generated expression signatures by comparing bulk RNA-seq profiles between AD patients and elderly controls with neurodegenerative diseases from the Mayo RNAseq Study (syn5550404, AD Knowledge Portal), which includes whole-transcriptome data for 275 cerebellum (CBE) and 276 temporal cortex (TCX) samples. The temporal cortex is heavily affected by tau and amyloid pathology, whereas the cerebellum is relatively less impacted by AD and often serves as an internal control tissue[29]. In addition, we compiled 95 drugs that have been investigated or approved for AD from ALZFORUM and are predicted to cross the blood-brain barrier according to SWISSADME, along with 1,000 randomly selected FDA-approved small molecules. These compounds were subjected to Z-RGES computation. As expected, AD drugs showed stronger reversal of TCX expression signatures than other FDA-approved drugs, whereas this pattern was not as evident for CBE (Figure S2D). Furthermore, using the TCX expression profile, we screened the ZINC library and found that AD drugs were highly enriched on the left side of the distribution, suggesting that, compared to library compounds, AD drugs are more likely to reverse AD-associated gene expression (Figure S2E). Moreover, some novel compounds demonstrated superior scores, indicating potential candidates for further hit validation.

The exploratory analyses using published data demonstrate that GPS predictions correlate with drug efficacy in multiple diseases and that the strength of this correlation depends on disease signatures which vary across data resources and parameters. Similar to target identification in traditional target-based approaches, disease signature identification is critical in the transcriptomics-based approach. Guided by this insight, we intiatiated a pilot study to optimize niclosamide, an anti-HCC hit compound identified in our previous drug repurposing study, which failed to advance due to poor water solubility and a narrow therapeutic window, while the disease signature was robustly validated (Figure S3A).

Seven commercially available niclosamide analogs were initially selected from MilliporeSigma by a chemist. Their anti-HCC Z-RGES scores were found to correlate with their IC_{50} values in HepG2 and Huh7 cell lines (Figure 3F). We then searched the virtual ZINC library for more niclosamide analogs with greater water solubility (predicted with ADMETlab 2.0[30]) and similar Z-RGES (Figure 3G and S3B, and Data S7). One analog Cpd.5338385 exhibited the highest potential for reversing the HCC signature (Z-RGES =

−8.13) and achieved sub-micromolar IC_{50} values in three HCC cell lines, but it unexpectedly induced hepatocyte growth (Figure 3G and Figure S3C). Another candidate Cpd.5260420 showed improved solubility (kinetic solubility in 1x PBS, Figure S3D) and no hepatocyte toxicity although it was less potent than niclosamide on Huh7 cells (Figure 3G and Figure S3C). Thus, the latter Cpd.5260420 was further evaluated *in vivo*. When administered intratumorally at 1 μg every three days for two weeks, Cpd.5260420 significantly reduced tumor volumes in the treated Huh7-xenografted mice compared to vehicle-treated mice (Figure 3H). The validation of this compound's activity supports the reliability of the GPS platform for drug discovery and lead optimization.

De novo design and mechanism of action elucidation of anti-HCC compounds from GPS-predicted gene expression signatures

In addition to discovering improved niclosamide anologs, we sought novel compounds for HCC therapeutics. Using the HCC signature, we screened seven million drug-like and in-stock compounds from the ZINC database. Z-RGES ranged from −10 to +2, with a distribution similar to that of the PRISM compounds (Figure S4A). To avoid non-selective compounds, we penalized compound profiles with undesired gene expression effects (e.g., up-regulation of over expressed genes in HCC) (Figure S4B). Eighteen structurally diverse candidates with high potential for reversing the HCC signature and low structural similarity to known anti-HCC compounds in ChEMBL were tested for inhibitory activity in three HCC cell lines, HepG2, Huh7, and Hep3B (Figure 4A and B, Figure S4C and D, and Data S8). One third of the proposed compounds showed significant inhibition across all three cell lines (Figure S4D), in concordance with our preliminary estimation of a hit rate of 40%. The top scoring candidate, 44443110, achieved IC_{50} values of 2~3 μM (Figure 4C), comparable to that of niclosamide (Figure S3A), suggesting that our pipeline can discover effective drug hits from a large virtual library generated by GPS. Another promising hit, PB56874852, with IC_{50} values ~4 μM, did not affect the viability of normal primary hepatocytes even at a high concentration of 100 μM (Figure 4D), suggesting enhanced selectivity over niclosamide. We further profiled the RNA-seq of Huh7 cells treated with DMSO (control), or with PB56874852 at 4 μM or 10 μM. The control group and the 4 μM group clustered closely, whereas the 10 μM group formed a distinct cluster (Figure S4E). The GPS predicted differentially expressed (DE) genes showed a similar pattern to the RNA-seq data for the 10 μM treatment of PB56874852 (a concentration used to train GPS, Figure 4E), and no correlation with that of 4 μM (Figure S4F), which presented too few DE genes. Interestingly, compared with other chemicals effective in Huh7 cells reported in literature (IC_{50} from 2 to 8 μM, ChEMBL v35), both hits showed good efficacy with lower molecular weights (318 and 379) than the average value 463. A similar compound among the top candidates was also found to selectively inhibit HCC cell lines' viability with low μM IC_{50} values (Figure S4G), suggesting the potential of this scaffold for hit-to-lead optimization.

Because of the considerable efficacy and high selectivity of PB56874852, we next implemented MolSearch to aid its optimization. MolSearch is a two-stage multi-objective hit-to-lead optimization algorithm based on Monte Carlo tree search. The goal was to design novel analogs of PB56874852 that achieve a better Z-RGES against HCC and retain favorable medicinal chemistry properties. Several proposed derivatives presented enhanced

HCC signature reversal by replacing the furan with halobenzene moieties (Figure 4F). Figure 4G shows our molecular design scaffold and the resulting IC_{50}s. In accordance with the MolSearch recommendation, the IC_{50} on Huh7 cells dropped to 0.34 μM when the furan was replaced with para-bromobenzene, but no significant change was observed when replaced with benzene, pyridine, or methoxy groups (Figures 4G and S4H). Further substitution with the electron-withdrawing trifluoromethyl group optimized the IC_{50}s to the sub-micro molar level for three HCC cell lines (MSU-45302 in Figure 4G), demonstrating several-fold greater activity than the first-line target therapy, sorafenib (Figure S5A). Under the oral dosing of 100 mg/kg, C_{max} and $t_{1/2}$ were 29 μmol/L and 2.5 h, respectively, and the plasma exposure data suggests that it can be maintained at 1 μM over 5–10 h (Figure S5B and S5C). Treatment of subcutaneous Huh7 xenografts in mice with intratumor injections of 1 μg MSU45302 every three days for two weeks resulted in significant reductions in tumor volumes over that of vehicle-treated mice (Figure 4H).

Compared to traditional phenotypic screening, GPS provides rich transcriptional information relating to each compound, facilitating the elucidation of mechanisms. Of note, the abovementioned calculation of transcriptome reversal captures phenotypic correlations but does not provide mechanistic insights. To uncover the mechanism of action (MoA), we investigated 26 anti-HCC compounds with different chemical moieties (including 15 analogs of MSU45302) by clustering their GPS-generated gene expression signatures (Data S10). As expected, compounds with moderate efficacy and belonging to the same chemical type presented similar transcriptomic perturbation signatures (e.g., series 1a and 2a in Figure 5A); whereas compounds showing high efficacy clustered together, although their scaffolds differed (e.g., niclosamide and MSU45302 in Figure 5A). This observation prompted us to explore the SGAR of MSU45302 (Figure 5B). Transcriptome-wide association analysis revealed that the anti-HCC efficacy of the 26 compounds significantly correlated with expression change of 15 genes including WDR75, KIF23, UHRF1, and MCM6 (Figure 5C). Such changes were further confirmed by the RNA-seq data from the xenograft models treated with MSU45302 and Huh7 cells treated with PB56874852 (Figures 5D and E). Among those genes, UHRF1 exhibited the most significant expression change (Figures 5D and E), and its knockdown significantly reduced HCC cell viability (Figure 5F and Figure S5D). We further confirmed the reduction of UHRF1 protein expression after MSU45302 treatment in Huh7 and HepG2 cells at 5 and 10 μM, respectively (Figure 5G and Figure S5E).

Moreover, patient survival analysis suggested UHRF1 as an unfavorable prognostic marker in liver cancer (Figure 5H), and spatial transcriptomics revealed widespread expression of UHRF1 in HCC tissues compared to controls (Figure 5I). UHRF1 encodes a member of a subfamily of RING-finger type E3 ubiquitin ligases, and is a critical regulator of DNA methylation[31,32]. Our data suggest that MSU45302 may inhibit HCC tumor growth partly *via* the suppression of UHRF1 and its downstream DNA hypomethylation. Taken together, our validations support the use of GPS coupled with SGAR for lead optimization and mechanism elucidation.

Reversal of cell type-specific gene expression identified repurposing candidates and novel compounds for IPF.

IPF is a pathological condition resulting from injury to the lungs and an ensuing fibrotic response leading to thickening of the alveolar walls and the obliteration of the alveolar space. The etiology of IPF and the role of the microenvironment in disease progression are largely unknown[22]. We first used the repurposing pipeline to explore disease biology for the identification of relevant cell populations, and then applied the screening pipeline to discover novel compounds (Figure S6A). Using bulk and single-cell RNAseq data from multiple datasets, we constructed signatures that capture different aspects of IPF biology, such as those related to cell type-specific transcriptional changes and those associated with disease development in animal models. We predicted and tested drugs targeting individual and combined signatures and then selected those that yielded positive drug hits in *ex-vivo* models. We eventually prioritized two signatures, one derived from epithelial cells and the other from mesenchymal cells, for screening of novel compounds from the Enamine HTS library. Top compounds were further evaluated in human precision-cut lung slices (PCLS) models.

To account for the diversity of cell types that might be involved in IPF pathogenesis, we first incorporated a large singe-cell RNAseq dataset[33], which contains single-cell transcriptomics profiles of lung samples collected from 10 control subjects and 12 patients with IPF (Figure 6A). We created signatures by comparing diseased samples and controls (Data S11), which revealed profound changes in the transcriptional features of all major cell types, including endothelial, epithelial, immune, and mesenchymal cells (Figure 6A). Additionally, we created signatures by comparing transitional alveolar type 2 (AT2) to AT1 cells and $KRT5^-/KRT17^+$ cells to AT1 cells in IPF patients based on previous studies[34,35]. In addition to the single cell data, we also used a set of signatures derived from bulk RNAseq performed in a bleomycin-induced fibrosis mouse model[36]. Finally, we incorporated a human bulk RNA-seq dataset[37] into our analysis, resulting in a set of disease signatures of potential interest.

Using the OCTAD pipeline, we predicted and selected 20 drugs for testing in mouse PCLS models (Figure 6B, Data S12). Drugs that reduced marker expression in the majority of tissue slices were further evaluated in human PCLS models. Among the four selected candidates, pyrithyldione consistently reduced fibrosis markers across PCLS models from eight patients, showing efficacy comparable to one of the FDA approved drugs nintedanib (Figure 6C, S6B). We further observed an anti-fibrotic effect of pyrithyldione after two-week treatment in the Bleomycin (BLM)-induced lung fibrosis mouse model (Figure 6D).

As pyrithyldione may cause life-threatening side effects[38], advancing it to clinical trials for IPF will require further optimization and evaluation. Nevertheless, the positive preclinical data prompted us to leverage pyrithyldione as a way to gain deeper insight into IPF pathogenesis which could guide the discovery of novel compounds. To explore its effect in the tissue microenvironment, we next performed bulk RNAseq on PCLS with the treatment of pyrithyldione. Deconvolution of these bulk RNAseq samples[33] with CIBERSORTx revealed a significant reduction in the myofibroblast fraction (Figure 6E, S6C, Data S13), consistent with the marker expression observed in qPCR and Western Blot analyses (Figure

6C). Furthermore, myofibroblasts being a potential target cell population aligns with the gene expression reversal prediction (Figure 6B). In addition to myofibroblasts, pyrithyldione showed the strongest reversal of the MUC5B$^+$ epithelial cell signature (Figure 6B).

To identify the genes for which expression is reversed by treatment of pyrithyldione in the RNAseq data, we mapped the disease signatures onto the bulk RNAseq data. We created a difference matrix by subtracting the log2 normalized expression values of untreated samples from treated samples for each patient, and then calculated the mean difference for each gene. We then selected significant DE genes of each signature and identified those with opposite sign in the mean expression data. Several genes in myofibroblasts were reversed while many more genes in MUC5B$^+$ epithelial cells were reversed, reinforcing the significance to target epithethials in IPF. GO-term enrichment analysis of these genes revealed that iron ion transport pathway and transition metal ion transport pathway are involved (Data S14), suggesting an important role of iron metabolism in IPF progression, which is supported by previous studies[39] and a potential mechanism by which pyrithyldione reverses fibrosis.

We further selected these two cell type signatures, representing epithelial and mesenchymal cell populations, as our main signatures of interest in novel compound screening. Using the GPS pipeline, we screened the Enamine HTS library for all the signatures and selected the top 40 compounds predicted to reverse the myofibroblast and the MUC5B$^+$ epithelial signatures. These compounds were then analyzed and only those with favorable medicinal chemistry properties (structural liabilities, mutagenicity and cLogP) were selected, resulting in 19 compounds that advanced to testing (example compounds shown in Figure 6G). Among these, four of them showed a reduction in fibrotic markers in one patient PCLS (Figure 6G, S6D) and drug 18 showed consistent statistically significant reductions in FN1, SMA and CTHRC1 across multiple samples (Figure 6H). The identification of the repurposing candidate and the novel compounds demonstrated the potential of our transcriptomics-based approach for drug discovery in IPF.

Discussion

The advances of transcriptomics technologies have led to the accumulation of molecular features for diseases, providing a tremendous opportunity for drug discovery. The two main drug screening approaches – target-based and phenotypic screening[40] – were developed before the emergence of omics technologies; not surprisingly, neither is prepared to fully utilize the previously unanticipated voluminous omics data. The target-based approach aims to discover a magic bullet from thousands of features; however, many diseases are driven by multiple factors rather than one. In HCC, none of the genomic features are altered in over 50% of patients, and none of the clinical targets are highly expressed in over 20% of patients[41]. In IPF, distal lung epithelial cells have been implicated in its pathogenesis[42–44]. Recent studies have identified several new relevant cell populations in the distal lung epithelia in lPF lungs, such as basaloid (KRT5$^-$/KRT17$^+$) cells[33,45], alveolar-basal intermediate (ABI) cells[46], and cells in pre-alveolar type-1 (AT1) transitional state (PATS)[47]. These diverse related cell populations suggested that targeting different pathways across multiple populations is likely necessary for developing more effective therapeutics in IPF. Besides, the inherent challenges of the target-based approach are highlighted by the fact that small molecule drugs tend to target multiple proteins, and an off-target effect is a common mechanism of action of cancer drugs under clinical trials[48]. Although the phenotypic approach circumvents concerns about the off-target effects, little knowledge of the primary target poses challenges in lead optimization[2,6].

Using HCC and IPF as case studies, we have demonstrated the potential of applying a transcriptomics-based approach as a new means of discovering new therapeutic hits in cancer and non-cancer diseases. We have also demonstrated the power of leveraging single cell transcriptomics in IPF drug discovery. Compared to prior docking-based screening studies[4,49], this work leveraged high-dimensional transcriptomic features in novel compound screening. Compared to traditional phenotypic screening approaches[2,3], the rich transcriptomic features provide insights into drug mechanisms of action. Compared to numerous AI and machine learning guided drug design studies[1], this work performed down-stream compound synthesis, optimization, and validation, a process that typically takes years to complete. We have demonstrated that this approach not only leads to the discovery of novel compounds targeting distinct cell populations that contribute to disease progression, but also enables probing disease biology using emerging single cell RNA-seq and *ex-vivo* models.

Although our platform has enabled the discovery of novel lead compounds, direct comparison with first-line treatments and more extensive mechanistic studies of these compounds are still required. For example, in IPF, potential resistance mechanisms of the drug candidates in human lung microenvironments remain unclear. The concept of stromal cell-mediated drug tolerance is well-established in oncology, where many of the FDA-approved agents have only limited therapeutic benefit[50]. Similarly, the lung microenvironment in IPF can foster drug-tolerant states analogous to those seen in cancer. Excessive extracellular matrix deposition and remodeling not only stiffen lung architecture but can also impede drug penetration into critical cellular niches, thereby reducing therapeutic exposure[51]. To address this limitation, we adopted inhalation-based drug delivery to enhance local therapeutic concentrations while minimizing systemic side effects, as demonstrated in our preclinical models. Additionally, conventional target-based therapies typically focus on a single molecular pathway or cell type. For example, integrin inhibitors targeting $\alpha v\beta 6$ or $\alpha v\beta 1/\beta 8$ can partially reduce TGF-β activation, but latent TGF-β may still be activated by other αv integrins, thrombospondin, or proteases[52]. In contrast, our approach leverages the reversal of global transcriptomic features; the repurposed drug we identified can simultaneously reverse pathogenic signatures in both myofibroblasts and $MUC5B^+$ epithelial cells. This dual targeting strategy has the potential to overcome stromal cell-mediated drug tolerance and provide broader therapeutic efficacy in IPF.

Limitations of the study

First, although the LINCS database is the largest of its kind, the number of compounds for specific contexts (e.g., a given cell type under low-dose treatment) and the coverage of transcriptome (978 landmark genes) remain limited, thereby, restricting the generalizability of the model, the coverage of diverse chemical space and predictable genes. As a result, GPS currently predicts drug profiles across four cell lines and utilize their combined signatures in drug screening, as the Connectivity Map initially did[12]. As high-throughput transcriptomics become more advanced and cost-effective, and emerging machine learning models such as foundation models and few-shot learning enable knowledge transfer across multiple tasks and contexts[53], we anticipate that drug profiles in disease-specific models could be more precisely predicted in the near future, leading to more effective screening. Moreover, GPS aims to capture the underlying relations between transcriptional change and chemical structural features, rather than to uncover mechanistic causality. Incorporating uncertainty and model-interpretability would enhance model reliability and provide additional biological insights. Second, although SGAR provides insights into potential targets and pathways affected by a compound, it does not capture direct ligand-protein binding interactions. Structure-based modeling approaches including AlphaFold 3[54] may complement SGAR to elucidate mechanisms of action. Third, transcriptomic prediction alone is insufficient to capture the full spectrum of drug activities and side effects; therefore, integrating other omics modalities may provide a more comprehensive understanding of drug actions. Finally, while our system has shown the potential of using transcriptomics to identify new therapeutic candidates, the identified compounds require further preclinical validation, business assessment, and mechanistic studies for advancing to clinical trials, and application of our system to additional diseases will also need extensive preclinical validations. To encourage the community to adopt this therapeutic discovery strategy, we provide a web portal and Docker containers that facilitates the use of these tools.

Resource availability

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Bin Chen (chenbi12@msu.edu).

Materials availability

All unique/stable reagents generated in this study are available from the lead contact with a completed materials transfer agreement.

Data and code availability

RNA-seq data are deposited in the GEO under accession number: GSE291867, GSE291190, and GSE291833. The whole novel compound discovery platform GPS includes three components: (1) RCL: Robust collaborative learning to improve the quality of drug-induced gene expression signatures, (2) GPS4Drug: Predict drug-induced transcriptomic signatures based on chemical structures and run drug screening, and (3) Molsearch: multi-objective compound optimization guided by transcriptomic reversal. The code and data necessary to reproduce the results and implement the models are available through the zenodo (10.5281/zenodo.15198638)[55], and through the GitHub (https://github.com/Bin-Chen-Lab/GPS).

STAR Methods

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Culture of HCC cell lines: Human HCC cell line Hep3B and HepG2 were purchased from American Type Culture Collection (ATCC), whereas Huh7 was a gift from Dr. Mark Kay (Stanford University). Huh7 cells were cultured in Eagle's Minimum Essential Medium Medium (cat# 30–2002, ATCC); HepG2 and Hep3B cells in Dulbecco's Modified Eagle's Medium (cat# 11095098, Gibco). All media were supplemented with 10% fetal bovine serum (FBS), 100 U/mL penicillin, and 100 μg/mL streptomycin. Cells were maintained at 37°C in a humidified 5% CO_2 atmosphere.

HCC animal models: Animal work was approved by the Administrative Panel on Laboratory Animal Care at Stanford University, and carried out in compliance with all federal and local institutional rules for the conduct of animal experiments. All mice were housed under standardized conditions with a 12-hour light/12-hour dark cycle, maintained at 40–60% relative humidity and a temperature range of 20–24 °C. Mice were group-housed (no more than five per cage) with appropriate bedding and provided ad libitum access to food and water. To generate subcutaneous xenografts, 1×10^6 Huh7 cells were suspended in 100 μL of Dulbecco's PBS (Invitrogen); after mixing with 100 μL of Matrigel (Corning, NY), the mixture was injected subcutaneously near the right forelimb of 6-week-old male NOD scid gamma (NSG) mice (The Jackson Laboratory, Bar Harbor, ME). Male mice were selected to minimize hormonal variability associated with the estrous cycle and to reflect the male predominance of HCC.

IPF animal models: To investigate the therapeutic efficacy of pyrithyldione in pulmonary fibrosis, we used a Bleomycin (BLM) induced lung fibrosis mouse model. Only male mice were used in the current experiment due to their established sensitivity to BLM[36,56,58]. C57BL/6J mice were used in this study and were approximately 12 weeks old at the time experiments were performed. All animal experiments were conducted in compliance with institutional guidelines and were reviewed and approved by the Michigan State University Institutional Animal Care and Use Committee (IACUC) under protocol number PROTO202200396. Details for animal groups are provided in the METHOD DETAILS.

IPF lung explant samples and PCLS models: IPF lung explant samples were obtained from 14 patients, undergoing lung transplantation at Richard Devos Heart and Lung Transplantation Center, Corewell Health Butterworth Hospital, Grand Rapids, Michigan. Pathologic assessment confirmed the findings of advanced usual interstitial pneumonia (UIP) in the IPF subjects (Table S6). All protocols were performed in compliance with relevant ethical regulations approved by local Institutional Review Board (IRB #: CHW 2017–198); Written informed consent was obtained from all patients who participated in the current study.

A piece of the left upper lobe, where active fibrosis occurs, of the IPF lung explants were inflated using an appropriate volume of liquified low melting point agarose solution in physiological DMEM+F-12 media (1:1) supplemented with 0.5 % FBS, 20 μg/ ml Gentamycin, and 0.5 μg/ ml and Amphotracin B, prepared as 2% W/V solution. The inflated lungs were then cooled in ice to set the agarose and stiffen before slicing. PCLS were cut at a thickness of 400 μm with a vibrating tissue slicer (VF-510–0Z vibratome, Precisionary Instruments, San Jose, CA). Details for PCLS treatment groups are provided in the METHOD DETAILS.

METHOD DETAILS

Details of GPS model development

RCL framework: The RCL system consists of multiple neural networks, in which each network is an individual learner, while the remaining networks form a peer system[57]. The knowledge that each network uses to update its parameters is fused from all peer networks. The knowledge refers to data samples that are selected to calculate the gradients. In detail, each network aims to optimize the following objective:

$$min_{\boldsymbol{w}, \boldsymbol{v} \in [0,1]^n} E(\boldsymbol{w}, \boldsymbol{v}, \lambda) = \sum_{i=1}^{n} v_i L(y_i, f(x_i, w)) + g(v_i, \lambda)$$

where w is the model parameter, v is a latent variable associated with each training sample, and λ is a control parameter. $g(v, \lambda)$ is a function that regularizes the weight associated with each sample. A hard thresholding regularizer $g(v, \lambda) = -\lambda v$ yields the closed-form solution for $v^*(\lambda; l) = 1 \ if \ l < \lambda \ else \ 0$ at each iteration. The solution indicates that if a sample has larger loss on the current model, it is likely to be more difficult to learn or is even an outlier and will not be included for training at the next iteration. Therefore, such objective design can potentially mitigate the negative effect caused by noisy samples. To suit the mini-batch training strategy in deep neural networks, at each iteration, instead of using all data in a batch to train, only top $R(T) \times 100\%$ ranked small-loss samples will be selected to update the network parameters, where R is the utilization rate depending on training epoch T.

RCL further enhances the robustness by incorporating multiple networks, in which it first encourages disagreement between networks during the early training stage to mitigate sample selection bias of a single network learner and encourages agreement between networks during the later stage when the learners achieve certain accuracy. Specifically, consider each iteration, denote D_j as the samples selected by j-th network, i.e., $D_j = argmin_{D: |D| \le nR(T)} l(f_{\theta_j}, D)$ where n is the batch size, $R(T)$ the utilization rate and $|\cdot|$ the size of batch of samples, θ_j the model parameters for network j and $l(\cdot)$ the loss function. Network j will not use D_j, but rather use decisions fused from all the remaining networks $\{1,2, \ldots, K\} \setminus j$ to update its model parameters. Specifically, let

$$D_I = \cap_{k=1, k \ne j}^{K} D_k, \quad D_U = \cup_{k=1, k \ne j}^{K} D_k$$

be the intersection and union of selected samples from all peer networks except j-th network. The fused samples are obtained by:

$$D_j' = D_I \cup randon\_sample(D_U \setminus D_I, r(T) \times |D_U \setminus D_I|)$$

where $r(T)$ is the fusion rate that controls the disagreement strength. The formula for $r(T)$ is given by:

$$r(T) = 1 - \left(\frac{T}{\alpha T_c}\right)^{\beta} \quad if\ T < \alpha T_c, else\ 0$$

where T_c denotes the switch epoch after which only $R \times 100\%$ samples will be selected by each network before the fusion step, α and β are hyper-parameters. α determines the lag of switch epoch compared to T_c and β controls the decay strength of disagreement. Smaller β corresponds to a faster transition from disagreement to agreement. Finally, network j uses D_j' to calculate the gradients and update its model parameters (formula). Each network updates itself in the similar way and after all the networks have been updated, they enter the next iteration and repeat the process. The networks in the final stage are expected to exert the optimal performance.

GPS training data: The LINCS Phase I data from Broad Institute[7] contains 1.3 million normalized profiles and 11,000 small molecule perturbagens covering more than 70 cell lines, most of which are cancer cell lines. For each cell line, gene expression change readings (i.e., Z-scores) for 978 genes are available for different drug profiles under different dose and time conditions. In this work, we focus on four commonly studied cell lines: MCF7, HEPG2, PC3 and VCAP, measured at dose 10 μM and at 24 h treatment timepoint. The code could be easily applied to model other cell lines.

Gene predictability: For each gene in each cell line, a random forest regression model was trained to predict its expression (Z-score) using the ECFP4 features (2048 bits) of the respective compound, in a 10-fold cross-validation manner. In each fold, two models were trained on the true data and randomly shuffled data, respectively. Then we compared the mean squared error (MSE) and Pearson R metrics between the true and random data trained. A gene was considered predictable if the MSE was significantly smaller for true data than random ($P < 0.05$, t-test) and Pearson $R > 0.3$. A cell line was selected if it was associated with more than 100 predictable genes.

Label quality assessment: Each drug profile comprising expression of 978 genes is repeatedly produced under the same condition for a different number of times, which results in multiple technical replicates for the same drug profile. Some replicates have consistent Z-scores while others do not, which indicates the variation of label quality across different drug profiles. In order to ensure a meaningful validation result, a high-quality test set is required for internal validation. Therefore, we estimated the relative quality of a drug profile and generated a unique Z-score profile for each drug. Specifically, denote $y_{ij} \in R^{1 \times 978}$ the Z-scores for 978 genes of drug i under the j-th replicate for a given dose and time. We first obtained the averaged Z-score across all replicates and calculated the correlation between each replicate and the averaged Z-score:

$$\underline{y_i} = \frac{1}{N_i} \sum_{j=1}^{J} y_{ij}, \quad r_{ij} = cor(y_{ij}, \underline{y_i})$$

We obtained the final unique Z-score by weighted average of the replicates based on their correlation with the average Z-score:

$$\underline{r_i} = \frac{1}{N_i} \sum_{j=1}^{J} r_{ij}, \quad y_i' = \sum_{j=0}^{J} \frac{r_{ij}}{\underline{r_i}} y_{ij}$$

where $\underline{r_i}$ is the quality and y_i' the final Z-score profile for drug *i*. A higher correlation $\underline{r_i}$ indicates relatively better quality of Z-scores of the drug *i*.

GPS features and labels: A tuple of drug profile and gene *(drug, gene)* corresponds to Z-score $y_{(d,g)}$ which quantifies how gene expression changes relative to reference controls. The input comprises two sets of features: drug features and gene features. For drug features, we used the ECFP4 fingerprint. Although we explored multiple embedding methods based on graph neural networks, we did not oberve any superiority over ECFP4. This observation is consistent with the literature[59]; therefore, we implemented ECFP4. For gene features, we used 1107 GO terms preprocessed by[14], which characterize the functional and biological properties of genes. We discretized the target $y_{(d, g)}$ into 3 classes, ie., $y_{(d, g)} < -1.5$ (down-regulate), $-1.5 \le y_{(d, g)} \le 1.5$ (no change), $y_{(d, g)} > 1.5$ (up-regulate).

GPS network architecture: The network architecture consisted of fully-connected layers in the shape of (2131, 128, 32, 3) neurons at each layer. Leaky ReLu activation functions with a slope of 0.01 and a dropout rate of 0.5 and Adam optimizer with a learning rate of 0.001 were used. The model converged fast on real data, thus we ran 25 epochs and used the average over last 5 epochs as the final performance.

Experiment set-up: Drug profiles with $r_{dp} > 0.7$ were considered high quality. For each cell line, to construct training and testing datasets, we randomly sampled 10% of the high-quality drug profiles to serve as the testing drug profiles. The left-over high-quality profiles, combined with low-quality drug profiles, were used as the training drug profiles. Each drug profile was associated with a certain number of selected genes (Figure S1B) according to gene predictability. Since drugs capable of regulating most genes are rare, the labeled data was highly imbalanced. Therefore, we down-sampled the dominant class during training while keeping test set unchanged. We also added macro-f1 score as an evaluation metric alongside accuracy. In real experiments, the noise rate ϵ is unknown and thus treated as a hyper-parameter to tune. In order to ensure that all methods eventually include the same number of data points, we first searched across all possible noise rates {0.1, 0.2, 0.3,.., 0.9} for one network and picked the best performing rate in RCL. We tuned K={3, 4, 5} and α = {2,3,4} and β = {2,3,4} for the proposed method. Each method was run five times with different random seeds and their performances was averaged as the final performance.

For each gene regulation by a compound, GPS predicts the probabilities of it being up-regulated, no-change, or down-regulated, adding up to 1. Three median values of these probabilities across the four cell models were taken as the merged probabilities of the three categories (i.e., up, none, and down). A final label of up- or down-regulation was assigned if the merged probability of the up- or down-class was greater than 0.95, otherwise no-change.

External validation data processing: Raw RNA-seq data was downloaded from https://trace.ncbi.nlm.nih.gov/Traces/sra/sra.cgi?study=DRP006006. Data were processed using RSEM 1.3.1 + STAR 2.6.1 pipeline. Log2 transformed TPM values (addition of pseudocount 1) were used as gene expression measures for analysis. The RNA-Seq processing code is available at GitHub (https://github.com/Bin-Chen-Lab/chenlab_toil). Log2 fold changes for each compound treatment were computed by comparing with DMSO treatment for each cell line, and then the median across different cell lines was taken for each compound. Only samples with drug treatment at 2~4 μM for 24 h were included. After merging expression values of all cell lines, an up-regulation was defined as drug-induced gene expression with log2 fold change (average) greater than 0.5, and a down-regulation was defined as log2 fold change (average) less than −0.5.

Compare multiple chemical representations: Pretrained transformers Uni-Mol[60], ChemBERTa[61], and MolT5[62] were used for drug embeddings. For internal validation, we followed the RCL work for dividing training set and test set, where all drugs in the test set were unseen during training. For external validation, we used the same training set and test set as shown in Figure 1B. Code implementation:

1. Uni-Mol (default 768 dimensional embeddings, https://github.com/deepmodeling/Uni-Mol). "UniMolRepr" function was used for generating representations. Hyperparameters: 'data_type': 'molecule', 'remove_hs': False, 'model_name': 'unimolv2', 'model_size': '84m'.

2. ChemBERTa (default 768 dimensional embeddings): Pretrained ChemBERTa tokenizer and model were downloaded from: https://huggingface.co/seyonec/ChemBERTa-zinc-base-v1. "RobertaTokenizer" and "RobertaModel" functions were used for generating representations with default built-in hyperparameters.

3. MolT5 (default 512 dimensional embeddings): Pretrained MolT5 tokenizer and model were downloaded from: https://huggingface.co/laituan245/molt5-small. "T5Tokenizer" and "T5EncoderModel" functions were used for generating representations with default built-in hyperparameters.

Biological relevance of GPS predictions

Gene ontology enrichment analysis: GO enrichment was calculated using Gorilla[63] with the selected 2198 genes as the target list and all evaluated 23,981 genes as the background. GO terms with a FDR q value < 0.001 and enrichment odds ratio > 1.5 were considered as enriched. For each of the categories "biological processes", "cellular components" and "molecular functions", the top five enriched terms were selected for illustration.

LINCS compound profile correlation and aggregation: The Spearman correlation between two individual profiles was calculated using 978 landmark genes (Z-scores of Level 5), either from the same compound across different cell lines or from different compounds within the same cell line. When aggregating a compound profile across multiple cell lines, the expression change of a specific gene was calculated as its mean Z-score divided by its standard deviation across Z-scores of the cell lines of interest (e.g., GPS selected cell lines or other cell lines profiled).

Chemical and transcriptional space analysis: Compound-target interaction activities were retrieved from ChEMBL (v. 31). Several canonical pathways and well-studied therapeutic targets were selected. For each target, compounds with activities (the pChEMBL values) greater than 95% of all were kept, and then 20 of them were randomly selected. Both chemical (FCFP4, 1024-bit) and gene expression (GPS-generated, merged from the four cell models, 2198-bit) features were calculated. Compound clustering was visualized in UMAP, where the distance metrics were set as “jaccard” and “cosine”, respectively. To quantify clustering, 15 neighbors of each compound were analyzed in both chemical and transcriptional space. We calculated the recalled neighbors with the same target and the neighbors modulating the same pathway but different targets.

Correlation analysis with shRNA profiles: We selected 20 therapeutic targets from diverse categories, i.e., transporters, epigenetic targets, enzymes, membrane receptors, and ion channels. For each target, compounds showing activities (the pChEMBL values) greater than 95% of all were retained, and then 20 of them were randomly selected, resulting in a total of 400 compounds. The gene expression signatures of these 400 compounds were imputed by GPS and merged across cell models. A similarity score was calculated between each compound profile and each shRNA profile of the compound target (from LINCS v. 2020, landmark and well-inferred genes included), resulting in a 20 × 20 similarity matrix. The similarity score was defined as the sum of −log10 transformed p-values for up- and down-regulated genes enrichment. For example, if a compound was predicted to up-regulate n genes, then we performed the rank sums test between the LINCS Z-scores of the n genes and Z-scores of other genes to compute the p-value. For each target, we compared the similarity scores between its inhibitors and non-inhibitors.

Z-RGES computation and comparison

Z-RGES computation: The computation of RGES was modified from the connectivity score developed in previous studies[8,10,12]. Unlike previous studies, the query gene set here is associated with the compound and the ranked profile corresponds to the disease. Genes were first ranked by their expression values in the disease signature i. An enrichment score (es) of each set of up- and down-regulated genes by a compound was computed based on their positions in the ranked disease signature. Let P be the total number of genes in the disease signature and let m be the total number of genes up- or down-regulated by the compound. First a vector V of positions (1…m) was constructed for each gene in the compound profile on the basis of the values from the disease signature. Those were sorted in ascending order such that $V(j)$ I here $j = 1,2, \ldots m$. Then, for each set of up- and down-regulated genes, we computed a_up, a_down, b_up and b_down using the formulae provided in the Supplementary Material in Lamb et al.[12] If $a\_up > b\_up$, we set enrichment score $es\_up = a\_up$, otherwise, $es\_up = -b\_up$. If $a\_down > b\_down$, we set enrichment score $es\_down = a\_dow$, otherwise, $es\_down = -b\_down$. *es_up* represents the absolute enrichment of an up-gene list in the disease signature and *es_down* represents the absolute enrichment of a down-gene list in the disease signature. As the absolute RGES is sensitive to the number of genes regulated by a compound, it was normalized based on a random distribution. In short, for a disease signature, we first sampled the background distributions of *es_up* or *es_down* values for 1500 times given a specific number of up- or down-regulated genes from the compound side. Then the absolute up or down enrichment of a new compound was normalized with this distribution using Z-transformation. The final reversal score was defined as the normalized *es_up* minus the normalized *es_dow*. In summary, the Z-transformation was performed to normalize the absolute RGES against a background distribution of random gene sets of identical size. This converts the score into a measure of statistical significance, mitigating bias towards compounds with few, non-specific transcriptional effects and improving comparability across compounds.

Z-RGES comparison: We compared GPS-derived Z-RGES with other similar approaches. GPS comprises two major computational components: (1) the prediction of compound-induced gene expression based on chemical structure (RCL framework) and (2) the ranking of compounds based on their ability to reverse disease-associated gene expression. Several computational methods have been developed for predicting compound-induced gene expression, although the input and output formats may vary. In GPS, we observed that certain genes are not predictable based on chemical structures. For example, changes in the expression of surface proteins involve complex biological processes, making it difficult for current training data and models to capture the relationship between chemical structure and expression change of surface proteins. To address this limitation, we converted the regression problem into a classification problem, predicting whether genes are upregulated or downregulated, and focused only on genes that are predictable. While we compared GPS performance with several baseline methods, many other published methods[14–20] were not directly comparable. Therefore, we evaluated their predictive performance in downstream tasks related to the prediction of compound activity.

In our previous study, we found that the reversal of disease-associated gene expression correlates with drug efficacy, providing strong evidence that the reversal score can be used to rank compounds[8]. Other studies have adopted a similar approach[15,17,18,20]. We assessed this correlation using the same dataset from our prior work[8]. Unlike our previous study, where reversal scores were computed based on experimental data, here we computed reversal scores based on predicted gene expressions. For DeepCE, we ran main_drug_repurposing.py against the HCC signature and extracted correlation scores from the script[15]. For DLEPS, we used its web portal, providing the structures of the test compounds along with the upregulated and downregulated genes in the HCC signature as input[20]. For TranSiGen, the code does not directly support the prediction of new structures[18]. However, the repository provides precomputed predictions for compounds related to pancreatic cancer. Using the provided structures and disease signatures, we computed Z-RGES and compared it with the AUC data from PRISM[28]. Likely due to the limited data quality in large pharmacogenomics datasets including PRISM[64], the correlation was not strong in either methods. The dataset used in our main analysis[8] was manually curated and combined from multiple studies in ChEMBL. It is important to note that a systematic comparison with other methods in the prediction of drug-induced expression needs standard benchmarks, which is lacking in the community.

Computational methods in the HCC study

Novel compound screening for HCC: The drug screening library was downloaded from the ZINC database in June, 2020[65]. Only the drug-like and in-stock subset was used, including 6,857,774 compounds. Then each compound SMILES was submitted to GPS to predict the effect on the predictable genes' expression. Before drug screening, a summarized profile library was prepared by merging the results of the four cell models (i.e., MCF7, HepG2, PC3, and VCAP). Up-/Down-regulation was called if the average predicted probability reached greater than 0.95. The HCC signature are adopted from our previous studies[8,11], and Z-RGES was computed for each compound in the library based on their reverasal to the disease signature. Top 2000 compounds were clustered and representative compounds were selected for futher testing.

Lead optimization with MolSearch: MolSearch is a two-stage multi-objective molecular optimization method based on Monte Carlo tree search (MCTS)[66]. It starts with existing molecules and modifies them towards desired ones. The modification (action space) consists of transformation rules derived from large compound libraries[67]. The entire search process consists of two stages: a HIT-MCTS phase that aims to improve biological properties (e.g., inhibition scores to proteins, and reversal of gene expression), and a LEAD-MCTS stage that focuses on non-biological properties (e.g., drug-likeness, synthetic accessibility, and solubility). Each stage contains a multi-objective Monte Carlo search tree where different property objectives are considered separately rather than combined. For each search tree, the search process consists of four steps:

a. Selection. Starting from the root node (lead compound), a best child is recursively selected from the Pareto node set calculated from the reward vectors until a leaf node, i.e., a node that has not been expanded or terminated is reached. The utility for each child is calculated as

$$U_k = \frac{R_k}{n_k} + \lambda\sqrt{\frac{4\ \ln n + \ln d}{2n_k}}$$

where R_k is the average reward obtained so far, n_k and n is the times child node k being visited and toal iteration. d is the reward dimension. The Pareto node set then can be computed given statistics of all child nodes. Once the set is computed, we randomly select one child in the set to proceed.

b. Expansion. The selected leaf node is expanded based on a policy until the maximum number of child nodes is reached. The possible actions are obtained using transformations derived from large compound libraries, which are chemically reasonable, covering a variety of modification directions, and being large in size in order to successfully navigate in the chemical space.

c. Simulation. From each child node, recursively generate the next state until termination and get the final reward. Along the way, we maintain a global pool of all Pareto molecules found so far. At each simulation round, given a termination state (molecule) with property score $S = (s_1, .., s_d) \in R^d$, by comparing it with all Pareto molecules in the global pool, the reward vector $R = (r_1, .., r_d) \in Rd$ of this state is defined as

$$r_i = \frac{1}{N_p} \sum_{l=1}^{N_p} \mathrm{I}\left[s_i > s_i^l\right], \forall i = 1, \ldots, d$$

where N_p is the number of Pareto molecules and s_i^l is the i-th property value of Pareto molecule l. We also update the global Pareto pool by adding new Pareto molecules if found and removing invalid ones based on the comparison result. The expansion policy in simulation is the same as b).

d. Backpropagation. The reward is backpropagated along the visited nodes to update their statistics until the root node using the following formula:

$$n_v \leftarrow n_v + 1, X_v \leftarrow X_v + R, v \leftarrow parent \text{ of } v$$

The two stages HIT-MCTS and LEAD-MCTS differ in how the candidate actions are picked given the original possible actions. In HIT-MCTS, the candidate actions are those yielding states with better property scores as compared to the current parent state. In LEAD-MCTS, the candidate actions are those producing states with better property scores than a constant threshold.

To find hit analogs for HCC, Z-RGES is used as the primary target score. The start compound is the lead compound PB56874852 and the desired candidates are those with lower Z-RGES. The naive HIT-MCTS does not yield diverse compounds with desired property improvement due to its imposing the same exploration-exploitation level at different tree depths. To mitigate the limitation, we introduce a temperature parameter t to control the exploration strength. The criteria of child nodes thus become:

$$s_c \geq t^{L - l_p} \cdot s_p$$

where L is the maximum tree depth, l_p is the parent depth. In our experiment, we set t=0.98, which indicates the strength of enforcing child's property score better than parent increases as depth increases, which encourages exploration at shallow layers while restricting explorative search at deeper layers.

After getting the candidates from HIT-MCTS, for simplicity, we then run LEAD-MCTS for the optimization of drug-likness and synthetic accessibility. The candidates from both stages are analyzed to guide the design of the final list for subsequent synthesis.

Quantify Gene-Structure-Activity Relationships: The gene expression signatures of selected compounds were generated using GPS, and their IC_{50} values in HCC cell lines (Huh7, HepG2, Hep3B) were determined. For each gene, we performed rank sums tests to evaluate whether the anti-HCC effect (the average IC_{50} of the three cell lines) was significantly associated with the regulation of this gene.

Spatial transcriptome analysis: The 10X Visium spatial transcriptomics data from five adjacent normal and five HCC tissue sections were obtained from Wu et al.[68]. All samples were processed using the STutility package (version 1.1.1). We generated a metadata table that included the locations of the corresponding files: filtered feature-barcode matrix (h5), spot file, H&E image file, JSON file, and tissue type. Quality control (QC) analysis was performed using the default criteria of STutility. The data were subsequently normalized using the SCTransform approach, and the normalized values were used for gene expression visualization.

Computational methods in the IPF study

Cell-type specific signature creation: Signatures were created from the single cell dataset[33] by using the Wilcoxon test available in the FindAllMarkers function in Seurat[69] to compare between IPF and Control patient samples for each cell population such as Mesenchymal, Immune, Epithelial and Endothelial cells. Additional signatures were created by comparing Transitional AT2 to AT1 cells and comparing KRT5-/KRT17+ cells to AT1 cells in IPF patients, as these cell types are pathological.

Quality control and signature selection: We excluded signatures with fewer than 40 genes and limited the DE analysis to the top 100 upregulated and downregulated differentially expressed genes. To assess the quality of the signatures derived from the single-cell dataset, we evaluated their ability to identify control drugs known to have some effect in IPF. The drugs selected were nintedanib and pirfenidone, the two FDA-approved drugs for IPF, camptothecin and its derivative hydroxycamptothecin[70], which have shown to inhibit fibrosis in other studies, as well as saracatinib, known to revert fibrogenic pathways and granted Orphan Drug Designation by the FDA[71].

We prioritized those cell type specific signatures that met our gene size requirements and in which at least one of the selected drugs had a sRGES score below −0.2, resulting in several preferred signatures including $MUC5B^+$, Mesothelial Cells, $MUC5AC^+$ High, Lymphatic Endothelial Cells and $SCGB3A2^+$ cells. Next, we assessed the quality of the signatures from the bleomycin induced IPF mouse model. The dataset included two batches with three signatures each: bleomycin-induced signature, a bleomycin-induced signature with ATP12A overexpression introduced by viral transfection, and a bleomycin-induced signature with GFP overexpression as a control for the transfection. ATP12A is a gene known to enhance fibrosis[36]. Among these, only two signatures (Bleo_Control and BleoAtp_Control_B9) met our criteria for gene size, and Bleo_Control recovered nintedanib as a top hit. In addition to the mouse dataset, we analyzed the cultured small airway bulk RNAseq dataset consisting of two signatures: normal lung tissue with and without TGF-β overexpression compared with fibrotic IPF lung tissue.

Drug repurposing and novel compound screening: Leveraging the unique patient-derived PCLS model, which preserves the tissue microenvironment, we first used efficacy data from a limited set of drugs to identify key cell populations relevant for therapeutic discovery. At this stage, we prioritized existing drugs because of their ready availability and the extensive literature supporting their use in both *in vitro* and *in vivo* studies. To select candidate probes, we applied the OCTAD repurposing pipeline. Although GPS could have been used directly, we considered OCTAD more appropriate in this context, as its predictions are grounded in experimental data, whereas GPS relies on computational models. Accordingly, we first used OCTAD to nominate a small set of probe candidates and then applied GPS to screen for novel compounds.

We ran drug predictions for each signature using the OCTAD R package and combined the results to identify the top 20 drugs that reverse the largest number of signatures. First, we ranked them based on the Bleo_Control mouse signature, which passed our quality control and was able to predict nintedanib as an effective drug and chose drugs that had a score below −0.2. We reasoned that this selection strategy would increase the likelihood of being effective in downstream validation in mouse models. We then selected drugs that had a score below −0.2 in all but two signatures that passed our quality control, resulting in a pool of 100 drugs. To refine the list further, we selected drugs that had predicted sRGES in SMALL_CONTROL_IvsH below −0.3, added them to the 100-drug pool, and removed drugs from the pool that had a SMALL_CONTROL_IvsH score above −0.3, which resulted in 64 drug candidates. This step was taken to enhance the likelihood of success in patient sample validation during downstream experiments. To include information from all datasets that met our gene size cutoff and determine the final 20 drugs for validation, we selected those 64 candidates and summed their rank across all interesting signatures. The top five drugs with the lowest rank sum were selected, allowing to incorporate additional information from signatures not considered in the initial analysis. Candidates specific to the Transitional AT2 to AT1 cells and KRT5-/KRT17+ cells to AT1 cells signatures as well as those for the signatures from the Mesenchymal and Epithelial cell populations were provided separately. For individual cell type signatures, we selected top five from each signature, while for the parental cell populations (e.g., epithelial cells), we ranked the drugs across all signatures corresponding to cell types within each parental population and selected those with the lowest rank sum.

For our novel compound discovery, we used our pipeline GPS to screen the Enamine HTS library, which contains more than a million novel in-stock compounds. Compounds with zRGES score less than −2 in the reversal of two interesting cell populations were considered hits.

Bulk RNAseq analysis of treatment samples: We performed RNA-seq on PCLS from three patients, both with and without drug treatment at 10 μM for 72 hours. The lung tissue slices were treated with pyrithyldione or DMSO. DE analysis was conducted by comparing each drug-treated sample to the control sample from the same patient. RNA sequencing data was processed using the in-house pipeline to generate transcript abundance[27]. The resulting count matrix was uploaded to the ExpressAnalyst online analysis tool, where EdgeR was used to perform DE analysis[27,72]. Genes with a p-value ≤ 0.01 and absolute log 2 Fold Change ≥ 1 were considered significant for all downstream analyses. Interpatient variation was considered in the design matrix of the edgeR analysis. Notably, DE genes were validated using qPCR, and their expression was mapped to single cells.

CIBERSORTx deconvolution: We performed deconvolution analysis using the CIBERSORTx pipeline[73] on the drug-treated and control bulk RNAseq samples. We used Docker image installation with default parameters and in single-cell mode. Individual patient single-cell data was used to construct references for deconvolution, and those patients leading to any significant cell types were chosen for final estimation. To determine the effect of the drugs on the fractional abundance of different cell types we compared the predicted fraction of each cell type before and after treatment using each single-cell signature matrix. The comparison was done using a paired t-test between the fractions of each cell type of the same patient before and after treatment.

Experimental methods in the HCC study

Cell proliferation assay: The effect of compounds on the growth of Huh7, HepG2, and Hep3B cells were assessed using the CellTiter 96 Aqueous One Solution Cell Proliferation assay (cat# G4100, Promega). Cells were seeded at 3×10^3 cells per well in 96-well plates and allowed to adhere overnight. Cells were then treated with compounds at a concentration range of 0.033 μM to 50 μM for 72 h, after which 3-(4,5-dimethylthiazol-2-yl)-5-(3-carboxymethoxyphenyl)-2-(4-sulfophenyl)-2H-tetrazolium (MTS) solution (15 μL) was added to each well for 4 h at room temperature. Stop Solution (100 μL) was then added to each well to solubilize formazan at 37°C for 1 h. Optical density (OD) values at 570 nm were measured using a SYNERGY-LX absorbance microplate reader (BioTek). The IC_{50} values were calculated through nonlinear curve fitting using GraphPad Prism 10.0 software.

Western blotting for UHRF1: HCC cells, HepG2 and Huh7, were treated with MSU45302 at 10x of its IC_{50} in each respective cell line. After 24 h treatment, total protein was extracted from cultured cells using the T-PER Tissue Protein Extraction Reagent (cat# 78510, Thermo Fisher Scientific), and quantified using the BCA Protein Assay Kit (cat# 23225, Pierce). Equal protein amounts (20 μg) were electrophoresed on 4–12% polyacrylamide gels (Invitrogen) and transferred onto polyvinylidene difluoride membranes. The membranes were then blocked using 5% nonfat milk for 1 h and incubated with primary antibodies against UHRF1 (cat# 12387S, Cell Signaling Technology, 1:1,000 dilution), GAPDH (cat# 60004–1-Ig, Proteintech, 1:20,000 dilution). Specific proteins were visualized utilizing a fluorescent secondary antibody, IRDYE 800CW goat anti-mouse (1:10,000 dilution, cat# 926–3221o, LI-COR Biotechnology) or IRDYE 680RD goat anti-mouse (1:10,000 dilution, cat# 926–68070, LI-COR Biotechnology).

UHRF1 knockdown: HepG2 cells (1.2 × 10^6) were seeded into 6-well plates and incubated overnight. For transfection, 100 pmol UHRF1 siRNA (Cat# s26555, Thermo Fisher Scientific) or negative control siRNA (Cat# 4390843, Thermo Fisher Scientific) was diluted in 150 μL Opti-MEM I medium. Separately, 18 μL Lipofectamine RNAiMAX Transfection Reagent (Cat# 13778150, Thermo Fisher Scientific) was diluted in 150 μL Opti-MEM I medium. The diluted siRNA and transfection reagent were then combined, incubated at room temperature for 5 min, and added dropwise to the corresponding wells. After 48 h of transfection, cells were harvested for cell viability assays and Western blot analysis.

Kinetic solubility measurements: A stock solution of known concentration of each tested analog was prepared (30 mM in 100 microliters of DMSO). The stocks were then serially diluted in a 384-well plate by a factor of three over six concentrations. The assay was run in triplicate starting with the 30 mM stock in the first row. The subsequent rows were stocked with 20 microliters of DMSO and 10 microliters of the solution from the previous row is added (with a robot-Beckman-Coulter Biomek FXP Laboratory Automation Workstation), serial dilutions repeated five times (30 mM, 10 mM, 3.3 mM, 1.1 mM, 0.36 mM). One microliter of these DMSO solutions were added to 99 microliters of 1x PBS in a 96-well plate, heated to 37°C for 2 hours, and read at 620 nm for absorbance. The data is processed in Prism™ and the kinetic solubility of each analog extrapolated from the graphs of absorbance vs concentration.

Pharmacokinetic studies: Top compounds (MSU-45301 and MSU-45302) tested for *in vivo* systemic exposure were completed by oral (PO) dosing to C57Bl/6 male mice (4 mice/group) at 100 mg/kg. MSU-45301 was dissolved in 95:5 DMSO/corn oil at 20 mg/mL active. MSU-45302 was dissolved in corn oil at 20 mg/mL active. All formulations were stored at room temperature, protected from light, until the time of use. Blood was collected at 30 min, 1, 2, 4, 8 and 24 hrs. Drug levels in plasma were measured by UPLC / MS / MS.

Xenograft mouse model and drug treatment: When xenografts become palpable, mice were randomized into two groups (n=5 each), and given intratumor injections of 50 μl DMSO (vehicle control), or 1 μg (dissolved in 50 μl DMSO) of compound 5260420 or MSU45302 every 3 days for 2 weeks. Tumor size was measured with a caliper before each treatment time point, and the tumor volumes were calculated using the formula: $V= 0.5 \times L \times \hat{W}2$.

RNA Sequencing of treated samples: Total RNA was extracted from cell lines treated with PB56874852, or tumor tissues mice treated with MSU45302, using RNeasy Plus Mini Kit (cat# 74136, Qiagen). Subsequent mRNA library preparation and sequencing, and data analysis were done by Novogene using their standardized protocols. For non-directional libraries, the process included end repair, A-tailing, adapter ligation, size selection, amplification, and purification, while directional libraries involved additional steps such as USER enzyme digestion. The constructed libraries were checked for their quality with Qubit, quantified by real-time PCR, and analyzed for size distribution using the Bioanalyzer. These libraries were then pooled based on effective concentration and data amount before being sequenced on Illumina platforms.

Experimental methods in the IPF study

Markers of fibrosis: Fibronectin, α-SMA, Collagen I, and CTHRC1 serve as crucial biomarkers in evaluating antifibrotic treatments. Their roles in fibroblast activation, ECM deposition, and disease progression provide valuable insights into therapeutic efficacy.

1. Fibronectin and α-SMA: Markers of Myofibroblast Differentiation. Fibronectin, particularly the ED-A isoform, and α-SMA are widely recognized markers of myofibroblast differentiation[74]. Myofibroblasts, which arise from fibroblasts in response to pro-fibrotic signals such as TGF-β, are the primary drivers of fibrosis[75]. Increased expression of these markers is associated with disease progression in fibrotic disorders[76].

2. Collagen I: A Major ECM Component. Collagen I is a principal structural protein in fibrotic tissue, and its excessive deposition contributes to tissue stiffness and organ dysfunction. Elevated Collagen I expression is a hallmark of fibrosis in various organs, including the lung, liver, and kidney[77]. Measuring Collagen I levels provides insight into the extent of ECM remodeling and fibrosis severity[78].

3. CTHRC1: A Novel Regulator of ECM Deposition. CTHRC1 is a secreted glycoprotein that regulates ECM turnover and tissue remodeling. It is a primary downstream target of TGF-β signaling and is specifically expressed by pathogenic fibroblasts exhibiting high collagen production[79]. Elevated CTHRC1 levels have been documented in IPF, systemic sclerosis, and post-COVID-19 lung fibrosis, correlating with disease severity. Additionally, CTHRC1 expression responds to antifibrotic therapies, making it a promising biomarker for treatment efficacy[80–83].

Preparation and treatment of PCLS: The slices were treated with 10 μM concentrations of pyrithyldione, bergenin, and nintedanib in DMEM+F-12 media (1:1) supplemented with 0.5 % FBS, 20 μg/ ml Gentamycin, and 0.5 μg/ ml and Amphotracin B, with corresponding DMSO-treated groups serving as controls. Treatments were performed under standard culture conditions for 72 h, with fresh drug solutions changed daily over the 3 days. PCLS were harvested and stored at −80°C until analysis.

Protein extraction and western blotting: IPF-PCLS were homogenized in RIPA lysis buffer (R0278, Millipore-Sigma, St. Louis, MO, USA) supplemented with protease inhibitors (A32955, Thermo Scientific, Waltham, MA, USA). Homogenates were centrifuged at 12,000 × g for 15 minutes at 4°C. The supernatant containing total protein was collected. Protein concentrations were quantified using the BCA assay (23235, Thermo Scientific) to ensure equal protein loading. Equal amounts of protein (5 μg for FN1 and SMA, 40 μg for COL1A1 and CTHRC1 per lane) were separated by 10% SDS-PAGE (NW04122BOX, Invitrogen, Waltham, MA, USA) and transferred onto PVDF membranes using the iBlot™ system (25 V, 6 minutes). Membranes were blocked with 5% non-fat milk in TBST (Tris-buffered saline with 0.1% Tween-20) for 1 h at room temperature to reduce non-specific binding. Membranes were incubated overnight at 4°C with the following primary antibodies: FN1 (1:1000, F3648, Millipore-Sigma, St. Louis, MO, USA), SMA (1:2000, A5228,Millipore-Sigma, St. Louis, MO, USA), COL1A1 (1:500,ABclonal, Cambridge, MA, USA), CTHRC1(1:500, 16534–1-AP; Proteintech, Chicago, USA), β-actin (1:2000, A2228, Millipore-Sigma, St. Louis, MO, USA) (loading control). Membranes were washed three times with TBST for 10 minutes each and then incubated with

HRP-conjugated secondary antibodies for 1 h at room temperature. Protein bands were visualized using a chemiluminescence substrate (34580, Thermo Scientific) and imaged with the ChemiDoc™ MP Imaging System (BioRad, Hercules, CA, USA). Protein band intensities were quantified using ImageJ software (NIH). The expression levels of FN1, SMA, COL1A1, and CTHRC1 were normalized to the expression of the housekeeping protein β-actin. Comparisons were made between drug-treated groups and their respective DMSO controls. Results are presented as fold changes relative to controls, with statistical analysis conducted to determine significance.

RNA isolation and bulk RNA sequencing: Lung slices were homogenized in TRIzol™ Reagent (15596026, Invitrogen, Waltham, MA, USA) and treated with chloroform to extract RNA. The resulting aqueous phase was transferred to a clean tube, and an equal volume of 70% ethanol was added. The mixture was then loaded onto a column for RNA extraction and purification using the RNeasy Mini Kit (74104, QIAGEN, Hilden, Germany), following the manufacturer's protocol. RNA concentration was measured using a NanoDrop™ OneC Microvolume UV-Vis Spectrophotometer (840274200, Thermo Scientific, Waltham, MA, USA). RNA samples were subsequently submitted to the Genomics Core at the Van Andel Institute for bulk RNA sequencing on an Illumina NovaSeq 6000.

In vivo validation of pyrithyldione: Mice were randomly assigned to one of three experimental groups: (1) untreated control group; (2) BLM group, which received a single dose of BLM; and (3) BLM + pyrithyldione group, which received a single dose of BLM followed by daily administration of pyrithyldione. There were at least 5 animals per group. BLM was administered on day 0 at a dose of 2 mg/kg body weight in 50 μL of sterile saline via intratracheal instillation. Pyrithyldione was administered at a concentration of 100 μM in 50 μL of saline once daily for 14 consecutive days via oropharyngeal aspiration. Mice were monitored daily throughout the study. On day 15, animals were euthanized by cervical dislocation under deep isoflurane anesthesia, and lung tissues were harvested for subsequent histopathological analyses.

Masson's Trichrome staining of mice lung tissue: Mouse lung samples were inflated with a 4% w/v paraformaldehyde (PFA) solution in phosphate-buffered saline (PBS) (14190250, Thermo Fisher) and then fixed in 4% PFA overnight. The samples were processed and examined histopathologically. Masson's trichrome staining was performed using a trichrome staining kit, following the manufacturer's protocol (ab150686, Abcam). The whole lung sections stained with Masson's trichrome were visualized, and images were acquired using a ZEISS Axioscan 7 slide scanner with a 20x objective. The acquired images were analyzed using NIH ImageJ software through color deconvolution and conducted in a blinded manner[5]. At least three lung sections with 10 random fields per section were analyzed for each experimental animal. Fibrosis index was calculated as the percentage of the lung tissue area occupied by collagen (blue) to the total lung tissue section area[36].

QUANTIFICATION AND STATISTICAL ANALYSIS

Figures from biological experiments were generated using GraphPad Prism and illustrations were created using BioRender, diagrams.net and BioIcons. Other figures were generated using R or Python. By default, statistical significance was assessed using the Student's t-test. For datasets that did not follow a normal distribution, the Wilcoxon rank-sum test was applied.

ADDITIONAL RESOURCES

Users can run compound-induced gene expression prediction and perform novel compound screening through the GPS web portal (http://apps.octad.org/GPS/), and perform drug repurposing through the OCTAD web portal (http://octad.org/) and the Bioconductor package octad.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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- Compound-induced transcriptomic changes predicted by GPS enable virtual drug screening
- Structure-Gene-Activity Relationship analysis elucidates drug mechanisms
- GPS identifies hits for HCC from a compound library and supports hit-to-lead optimization
- Transcriptomic reversal uncovers repurposing candidates and novel compounds for IPF

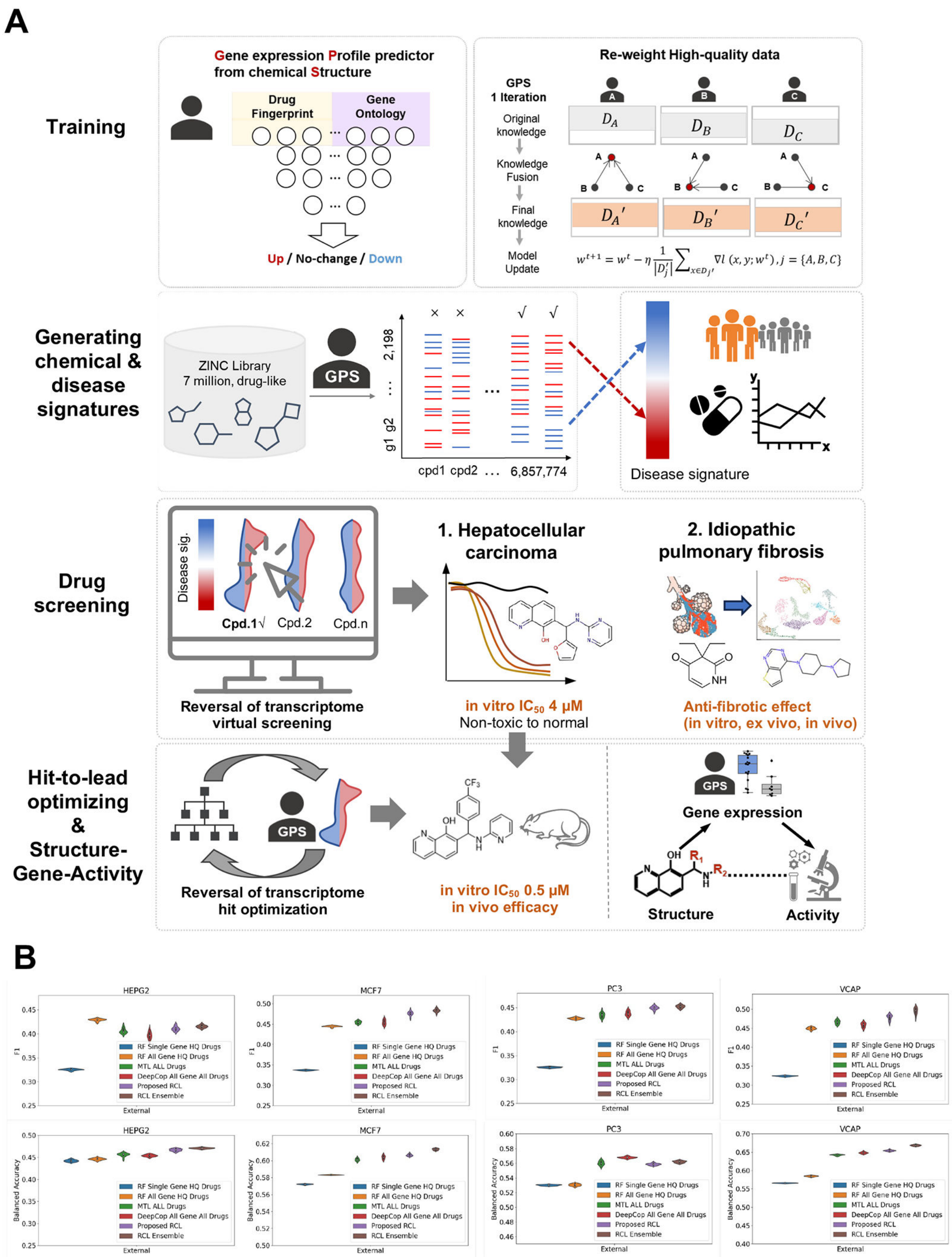

Figure 1. The GPS framework and performance.

A, An AI model for compound-induced transcriptomic signature prediction is developed using a robust collaborative learning strategy. This model enables the prediction of transcriptomic signatures of a large library of drug-like compounds based solely on chemical structures. The predicted profiles are ranked by their ability to reverse the expression of a signature derived from disease or phenotype profiles. Selected candidates are tested *in vitro* to identify lead compounds, which are subsequently optimized for improved reversal scores and favorable medicinal chemistry properties through a Monte Carlo tree search strategy.

Final candidates are evaluated in preclinical models and structure-gene-activity relationships are delineated for illustrating the mechanism of action. In this study, the ZINC library and the Enamine HTS library are used to screen compounds for HCC and IPF, respectively. **B**, The averaged F1 scores and balanced accuracies of 10 independent runs for HEPG2, MCF7, PC3, and VCAP under various comparison methods in an external validation. The tested drug-induced gene expression profiles are from LINCS 2020 at a concentration of 10 μM and treatment for 24 hours. Only high-quality profiles and those compounds excluded from the training set are considered. Note that in the ablation study, DeepCop was trained on all profiles without quality improvement, whereas RCL was trained using the same architecture and dataset but with corrected data. RF: Random Forest, HQ: High-quality, MTL: Multi-task learning, RCL: Robust collaborative learning, and SG: Single Gene. See also Figure S1.

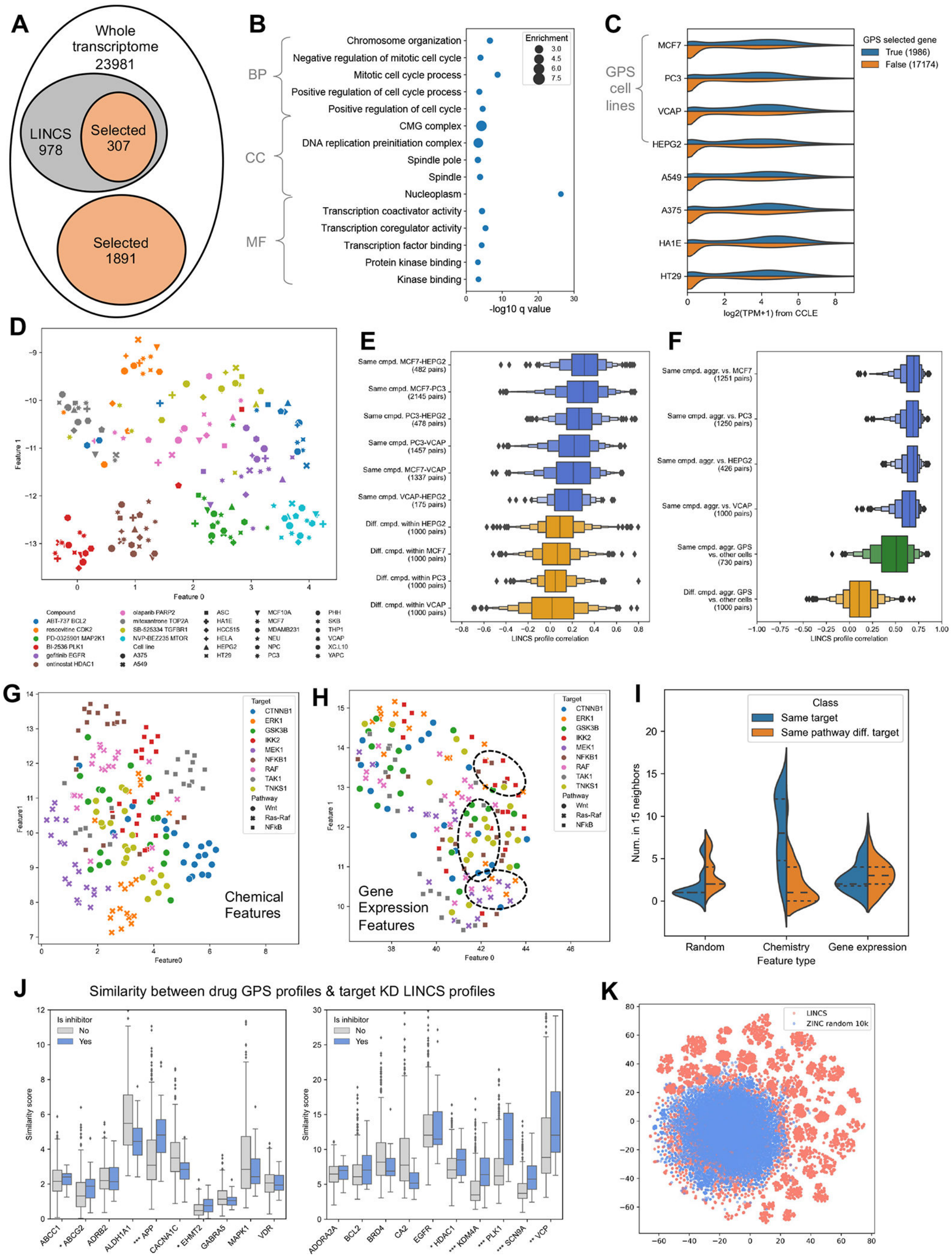

Figure 2. Biological insights of GPS-imputed compound transcriptomic signatures.

A, Number of GPS-selected genes that exhibit high predictability by GPS. **B**, Gene Ontology enrichment of the selected genes. **C**, Distribution of basal expression levels of selected vs. non-selected genes in untreated cell lines. **D**, UMAP projection of expression features for 10 example compounds profiled across several cell lines in LINCS. Each point represents a gene expression profile with colors denoting compounds and shapes indicating cell lines. **E**, Distribution of LINCS profile correlations for the same compound across different cell lines, compared to different compounds within the same cell line. **F**, Distribution of LINCS profile correlations of aggregated profiles vs. individual profiles across various cellular contexts, profiles aggregated from the four cell lines selected by GPS vs. from other cell lines, and aggregated profiles of different compounds. **G**, UMAP projection of inhibitors targeting nine proteins in three pathways using chemical features (ECFP6). Each point represents a compound, with colors denoting targets and shapes indicating pathways. **H**, Similar with **G** but using gene expression features predicted by GPS. Compounds modulating the same pathway but different protein targets are highlighted with circles. **I**, Violin plot showing the number of compound neighbors computed using chemical or transcriptional features. Blue indicates neighbors with the same target, and yellow indicates neighbors of the same pathway but different targets. **J**, Similarities between GPS-predicted compound profiles and shRNA profiles of their respective targets in LINCS. Each column denotes a therapeutic target. The Y-axis indicates the similarity scores. Blue and grey boxes represent the scores of inhibitors and non-binding compounds, respectively. *: $P < 0.05$, **: $P < 0.01$, ***: $P < 0.001$. **K**, t-SNE plot showing the distributions of compounds from LINCS and ZINC in chemical space.

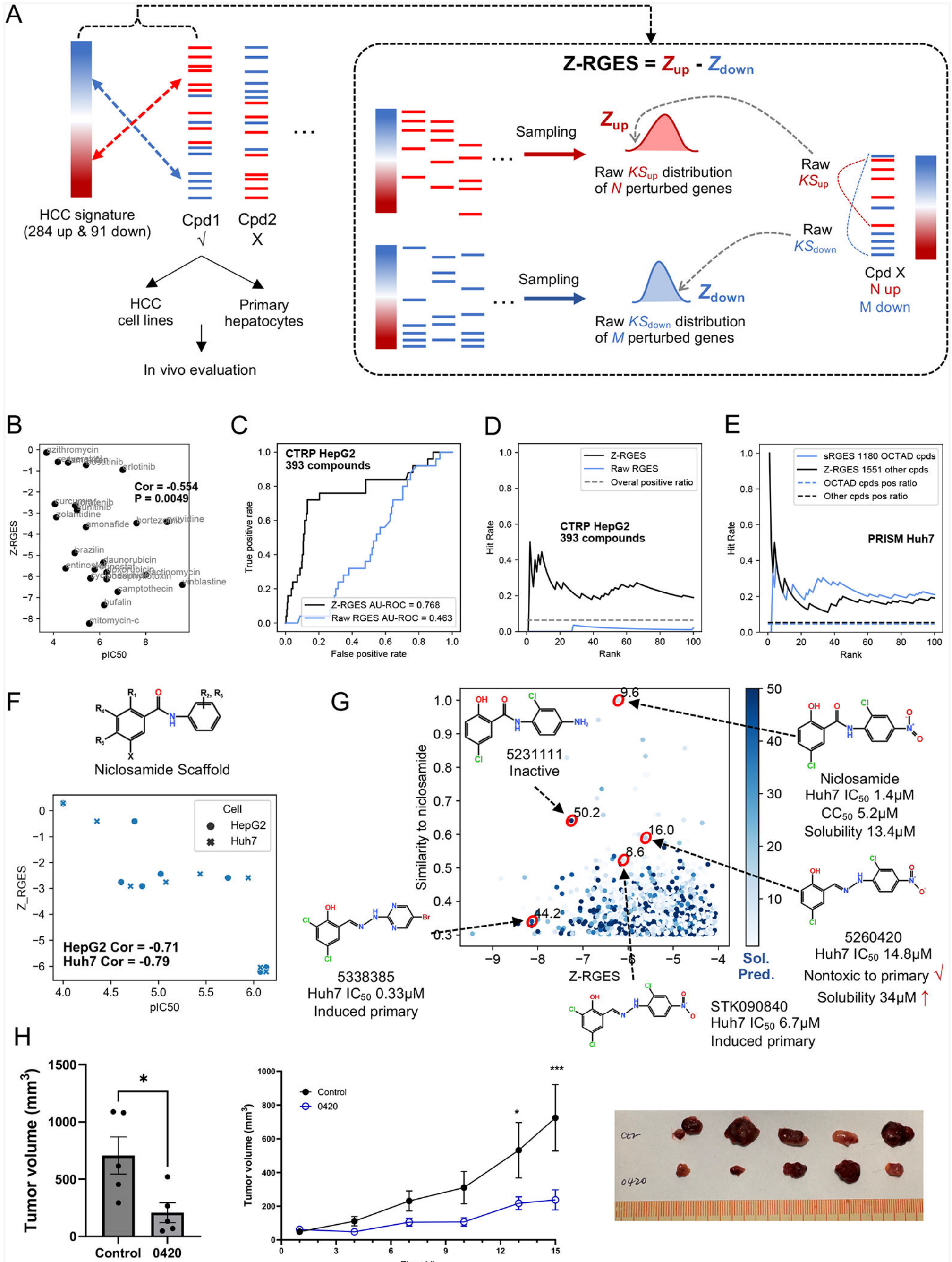

Figure 3. Validation of gene expression reversal using GPS-predicted compound profiles.
A, Illustration of the drug discovery workflow using a transcriptomics-based method, and an algorithm to calculate the reversal score Z-RGES using compound-induced and disease expression signatures. **B**, Correlation between Z-RGES and anti-cancer cellular activity of reported compounds. **C and D**, ROC curves (**C**) and hit rates (**D**) in using Z-RGES and raw RGES to rank compounds in one CTRP HepG2 dataset. **E**, Comparison of the hit rates between the published sRGES-based scoring using high quality drug profiles and the new Z-RGES-based scoring using predicted profiles. **F**, Scatter plot of Z-RGES for

niclosamide analogs and their IC_{50}s in HepG2 and Huh7 cell lines. **G**, Scatter plot of Z-RGES against the HCC signature and structural similarities to niclosamide. Niclosamide and four selected compounds are highlighted, with their validated activity, toxicity, and solubility. **H**: Anti-tumor effect of one analog, Cpd.5260420, in a subcutaneous Huh7 xenograft mouse model, showing significant reductions in tumor volumes after two weeks of intratumor administration. Two sided t-test was used (*: $P < 0.05$, **: $P < 0.01$, ***: $P < 0.001$). See also Figures S2 and S3.

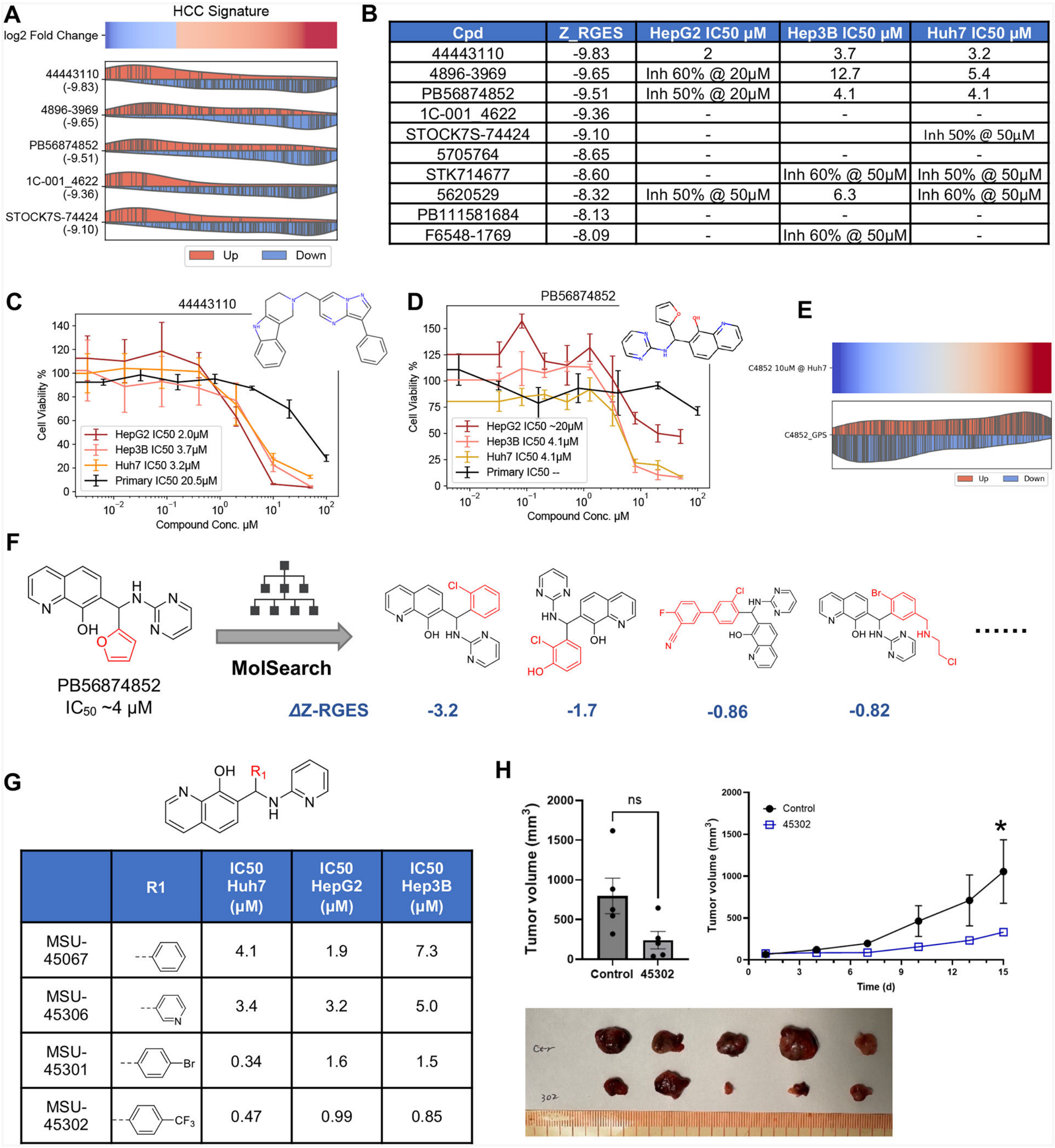

Figure 4. Large-scale screening for anti-HCC candidates and hit-to-lead optimization using GPS.
A, HCC signature reversal by the top five candidates selected from the ZINC library. Each bar indicates a gene up- (red) or down- (blue) regulated by a compound, aligning to the HCC signature. The compound's Z-RGES is shown in parentheses. **B**, Z-RGES and HCC cell line inhibition data of the top ten candidates. **C and D**, Dose-response curves of representative hits in HepG2, Hep3B, Huh7 and primary hepatocytes. **E**, comparison of the GPS-predicted profile and the RNA-seq data for PB56874852 in Huh7 cells. **F**, Representative analogs with better Z-RGES generated by MolSearch from PB56874852.

G, Representative designed compound structures and their inhibitory activities in HCC cell lines (IC_{50}s expressed in μM). **H**, Anti-tumor effect of MSU45302 in a subcutaneous Huh7 xenograft mouse model, showing significant reductions in tumor volumes after two weeks of intratumor administration (at 1 μg every three days for two weeks). Two sided t test was used (*: $P < 0.05$, **: $P < 0.01$, ***: $P < 0.001$). See also Figure S4.

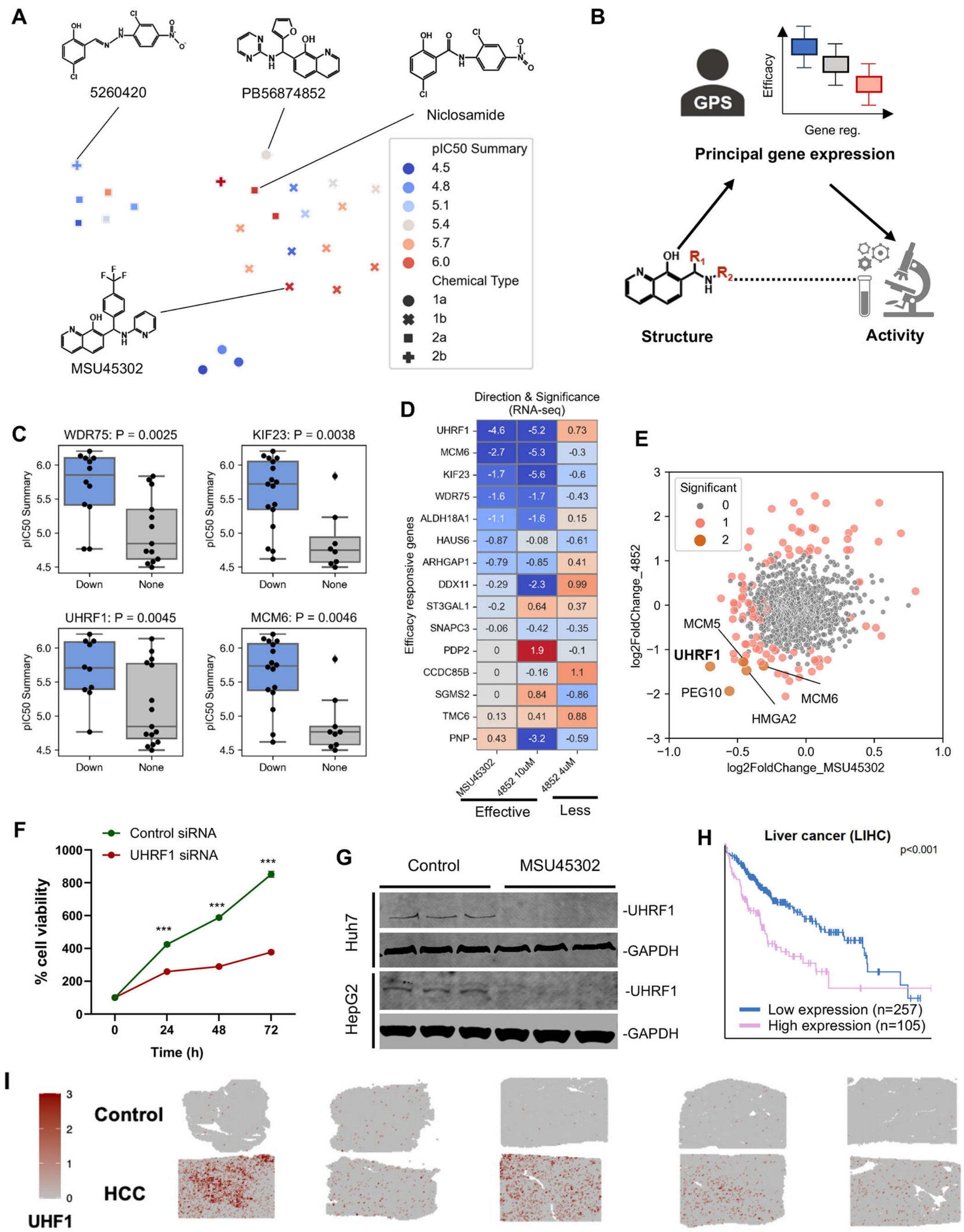

Figure 5. Structure-gene-activity relationship for HCC drug discovery and MoA interpretation.
A, UMAP projection of GPS predicted gene expression signatures for the anti-HCC compound series. Each dot represents a compound, with shapes indicating chemical series. Red indicates higher activity in HCC cell lines, while blue indicates lower activity. The pIC_{50} is the negative log of IC_{50} values in molar units. **B**, Illustration of the structure-gene-activity relationship (SGAR). **C**, Example genes whose expression changes predicted by GPS are associated with compound activity. **D**, Expression change of responsive genes after *in vivo* treatment with MSU45302 and *in vitro* treatment with PB56874852. Red: up-

regulation; blue: down-regulation. Values denote the logarithms of P-values from differential expression analysis. **E**, Log2 fold changes in the RNA-seq results for the two compounds. **F**, Viability of HepG2 cells after knockdown of UHRF1. **G**, Western blot showing UHRF1 protein expression induced by MSU45302 in Huh7 and HepG2 cell lines. **H**, Survival analysis of UHRF1 expression in liver cancer patients. Data was collected from the Human Protein Atlas website. **I**, The expression of UHRF1 in adjacent normal and HCC tissue sections using spatial transcriptomics. Red dots indicate spots with expression, while gray indicates no expression. The color scale represents the average normalized gene expression across spots within each tissue section. See also Figure S5.

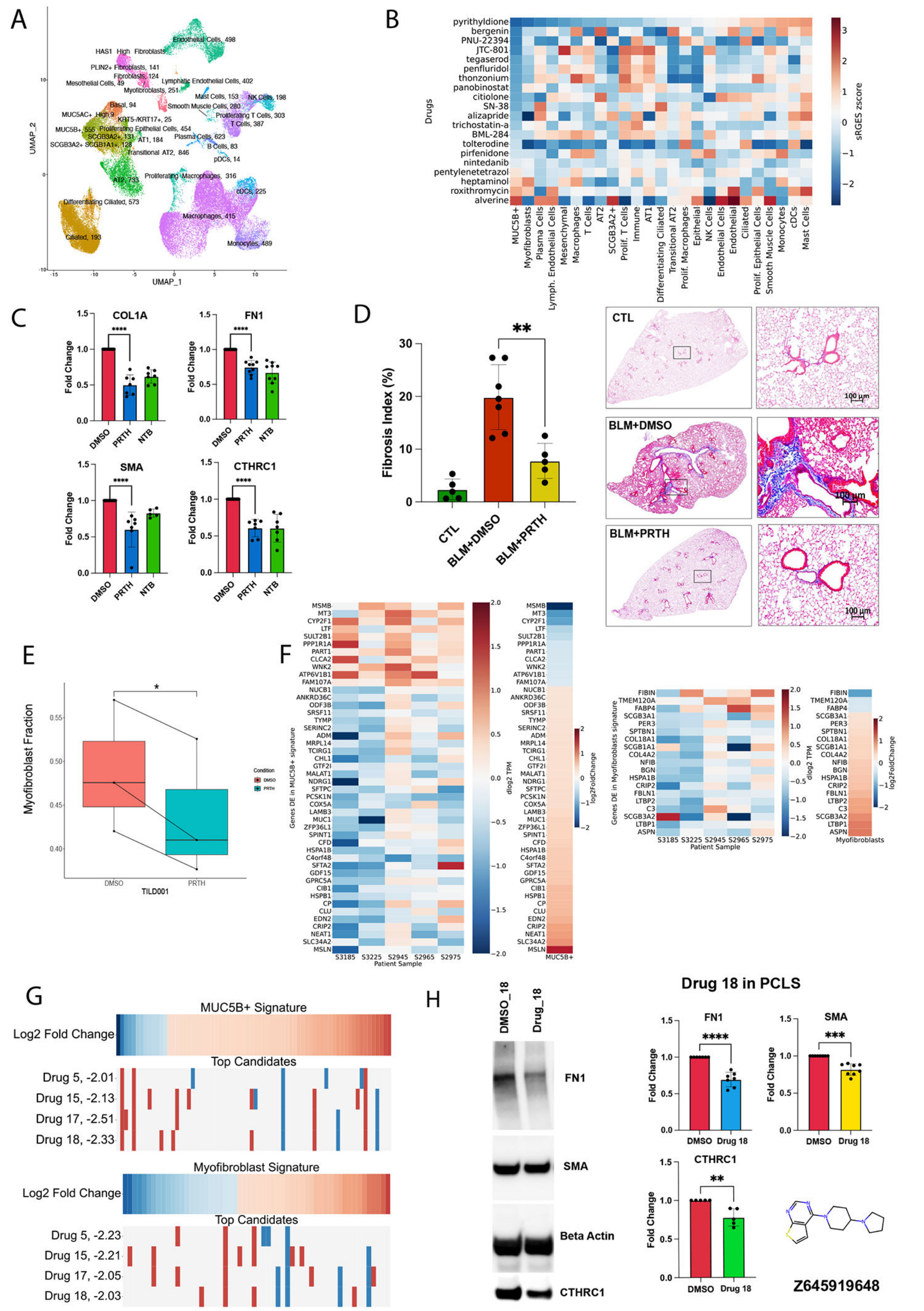

Figure 6: Reversal of cell-type specific signatures for IPF drug discovery.

A. UMAP of single cells from IPF patient samples showing the clustering of different cell types along with the number of their significant genes dysregulated in IPF (absolute log 2 fold change > 1 and adjust P < 0.05) (Adapted from[33]). The signature genes of each cell type were computed through the DE analysis using healthy lung scRNA-seq as control. For those cell types absent in healthy lungs such as KRT5^{-}/KRT17^{+} and Transitional AT2 cells, we computed the signatures by comparing to neighboring cell types within IPF patients.

B. Heatmap showing Z-normalized reversal scores between drug candidates and disease signatures that passed quality controls as described in Method section. Pyrithyldione which showed activity in PCLS models by inhibiting key pro-fibrotic markers is shown in the first row. **C**. Inhibition of profibrotic markers by pyrithyldione and nintedanib in human PCLS. We tested efficacy in *ex vivo* IPF lung slice models and measured protein expression for profibrotic biomarkers. We used the adjacent lung slices as control. Activity values were first normalized to beta-actin and converted into fold changes through the comparison between drug treated lung slices and adjacent lung slices with DMSO treatment. A t-test was performed between DMSO (normalized values) and treated samples. **D**. Anti-fibrotic effect of pyrithyldione in a bleomycin-induced fibrosis mouse model. Pyrithyldione attenuated BLEO-induced lung fibrosis *in vivo* (100 μM in 50μl saline, every day for 2 weeks through oropharyngeal aspiration). Representative bright-field microscope images of mouse lung tissue stained with Masson's trichrome show collagen deposition (blue) in the normal lungs, BLEO treated lungs, and pyrithyldione treated mice lungs. Collagen deposition area was calculated as % of total area. N=5 to 6. A t-test was performed between the BLEO vs the BLEO+ pyrithyldione group. **E**. Reduction of myofibroblasts in the bulk RNAseq of PCLS patient samples treated with or without pyrithyldione. Cell composition was estimated using CIBERSORTx. Each treated slice was matched with an adjacent control slice in a single patient and a paired t-test was performed. Three patients were used in total. **F**. Disease signature genes in $MUC5B^+$ epithelial cells and myofibroblasts reversed by pyrithyldione through the analysis of bulk RNAseq derived from PCLS with and without treatment. **G**. Novel compounds predicted to reverse the signature genes of MUC5B+ epithelial cells and myofibroblasts. Four novel compounds that exhibit anti-fibrotic activity in the PCLS model are shown. **H:** Anti-fibrotic effect of drug 18 across multiple fibrotic markers in PCLS. Drug 18, with a novel chemical scaffold, showed statistically significant reduction in FN1, SMA, and CTHRC1 in multiple patient samples at 10 μM. *: P < 0.05, **: P < 0.01, ***: P < 0.001, ****: P < 0.0001.See also Figure S6.

Key Resources Table

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
UHRF1	Cell signaling	Cat# 12387S, RRID:AB_2715501
GAPDH antibody	Proteintech	Cat# 60004-1-Ig, RRID:AB_2107436
IRDye 800CW Goat anti-Rabbit IgG	LI-COR Biosciences	Cat# 926-32211, RRID:AB_621843
IRDye 680RD Goat anti-Mouse IgG	LI-COR Biosciences	Cat# 926-68070, RRID:AB_10956588
IRDye 800CW Goat anti-Mouse IgG	LI-COR Biosciences	Cat# 926-32210, RRID:AB_621842
IRDye 680RD Goat anti-Rabbit IgG	LI-COR Biosciences	Cat# 926-68071, RRID:AB_10956166
FN1 (Anti-Fibronectin Antibody)	Millipore-Sigma, St. Louis, MO, USA	F3648
SMA (Anti-α-Smooth Muscle Actin (ACTA2) Antibody)	Millipore-Sigma, St. Louis, MO, USA	A5228
COL1A1 (Collagen I/COL1A1 Rabbit pAb)	ABclonal, Cambridge, MA, USA	A1352
CTHRC1 (CTHRC1 Polyclonal antibody)	Proteintech, Chicago, USA	16534
β-actin (Anti-β-Actin (ACTB) Antibody)	Millipore-Sigma, St. Louis, MO, USA	A2228
Bacterial and virus strains		
Biological samples		
Chemicals, peptides, and recombinant proteins		
Dimethylsulfoxid (DMSO)	SigmaAldrich	Cat# D2650
Niclosamide	SigmaAldrich	Cat# N3510-50G
Lipofectamine™ RNAiMAX Transfection Reagent	Thermo Fisher Scientific	Cat# 13778150
Pierce™ Protease Inhibitor Mini Tablets, EDTAfree	Thermo Fisher Scientific	Cat# A32955
Pierce™ Phosphatase Inhibitor Mini Tablets	Thermo Fisher Scientific	Cat# A32957
Phosphate buffered saline (PBS)	Gibco	Cat# 14-190-250
DMEM, high glucose	ATCC	Cat# 30-2002
MEM	Gibco	Cat# 11095098
Hepatocyte Cell Media	Sekisui XenoTech	Cat# K8200
Fetal bovine serum	Phoenix Scientific	Cat# ps-100-00-500
Cell banker 2	AMSBIO	Cat# NC0266420
Trypsin-EDTA (0.25%), phenol red	Gibco	Cat# 25200056
T-PER Tissue Protein Extraction Reagent	Thermo Fisher Scientific	Cat# 78510
Penicillin-Streptomycin Solution	Corning	Cat# 30002CI
Opti-MEM™ Reduced Serum Medium	Thermo Fisher Scientific	Cat# 31985070
Restore™ Fluorescent Western Blot Stripping Buffer	Thermo Fisher Scientific	Cat# 62300
PDI™ Povidone-Iodine Prep Pad	PDI	Cat# B40600
Alcohol Wipes	BD	Cat# 326895
ZINC000067803790	ChemBridge	44443110
ZINC000004631199	ChemDiv	4896-3969

REAGENT or RESOURCE	SOURCE	IDENTIFIER
ZINC000000086363	UkrOrgSynthesis	PB56874852
ZINC000100424631	Bionet	1C-001_4622
ZINC000169294501	IBSCREEN	STOCK7S-74424
ZINC000008743124	ChemBridge	5705764
ZINC000002361758	Vitas-M	STK714677
ZINC000100497241	ChemBridge	5620529
ZINC000004488711	UkrOrgSynthesis	PB111581684
ZINC000006738179	LifeChemicals	F6548-1769
ZINC000199299209	ChemBrdige	87967218
ZINC000080892226	UkrOrgSynthesis	PB1311104133
ZINC000015064122	ChemBridge	89237834
ZINC000253417069	Enamine	Z1950894777
ZINC000019889619	ChemDiv	Y511-5125
ZINC000004073497	Vitas-M	STL537005
ZINC000015223275	Vitas-M	STK005228
ZINC000426421820	ChemBridge	28984782
Pirfenidone	Selleckchem, Houston, TX,USA	S2907
Nintedanib Ethanesulfonate Salt	Selleckchem, Houston, TX,USA	S5234
Tegaserod (malate)	Cayman Chemical, Ann Arbor, MI, USA	14692
Roxithromycin	Cayman Chemical, Ann Arbor, MI, USA	19465
JTC 801	Cayman Chemical, Ann Arbor, MI, USA	21254
Wnt Agonist I	Cayman Chemical, Ann Arbor, MI, USA	19903
Pyrithyldione	Cayman Chemical, Ann Arbor, MI, USA	34528
SN-38	Cayman Chemical, Ann Arbor, MI, USA	15632
trichostatin A	Cayman Chemical, Ann Arbor, MI, USA	89730
Alizapride (hydrochloride)	Cayman Chemical, Ann Arbor, MI, USA	15645
Panobinostat	Cayman Chemical, Ann Arbor, MI, USA	13280
Thonzonium (bromide)	Cayman Chemical, Ann Arbor, MI, USA	31253
Alverine (Citrate)	Cayman Chemical, Ann Arbor, MI, USA	26892
Penfluridol	Cayman Chemical, Ann Arbor, MI, USA	26071
Bergenin	Cayman Chemical, Ann Arbor, MI, USA	26406
BiBF 1120	Cayman Chemical, Ann Arbor, MI, USA	11022
Heptaminol	Cayman Chemical, Ann Arbor, MI, USA	29879
Citiolone	Fisher Scientfic	50-187-4231
PNU 22394 hydrochloride	Millipore-Sigma	ADVH9B9B45E7
Tolterodine L tartrate	Millipore-Sigma	SML3034
Compound 3-1	Enamine US Inc. NJ, USA	Z1102413454
Compound 3-2	Enamine US Inc. NJ, USA	Z1233726600
Compound 3-3	Enamine US Inc. NJ, USA	Z1333156057

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Compound 3-4	Enamine US Inc. NJ, USA	Z135358136
Compound 3-5	Enamine US Inc. NJ, USA	Z1516314927
Compound 3-6	Enamine US Inc. NJ, USA	Z1562356677
Compound 3-7	Enamine US Inc. NJ, USA	Z1695946947
Compound 3-8	Enamine US Inc. NJ, USA	Z17375671
Compound 3-9	Enamine US Inc. NJ, USA	Z1747559808
Compound 3-10	Enamine US Inc. NJ, USA	Z1824576013
Compound 3-11	Enamine US Inc. NJ, USA	Z2047158994
Compound 3-12	Enamine US Inc. NJ, USA	Z2723076319
Compound 3-13	Enamine US Inc. NJ, USA	Z382984684
Compound 3-14	Enamine US Inc. NJ, USA	Z383196458
Compound 3-15	Enamine US Inc. NJ, USA	Z3888289046
Compound 3-16	Enamine US Inc. NJ, USA	Z52554493
Compound 3-17	Enamine US Inc. NJ, USA	Z57065103
Compound 3-18	Enamine US Inc. NJ, USA	Z645919648
Compound 3-19	Enamine US Inc. NJ, USA	Z98332604
Critical commercial assays		
MycoBlue® Mycoplasma Detector	Vazyme	Cat# D101-02
RNeasy Plus Mini Kit	Qiagen	Cat# 74136
4–12% NuPAGE BisTris Gels	Invitrogen	Cat# NP0329BOX
NuPAGE™ 4 to 12%, Bis-Tris, 1.0–1.5 mm, Mini Protein Gels	Invitrogen	Cat# NP0321BOX
NuPAGE™ 4 to 12%, Bis-Tris, 1.0–1.5 mm, Mini Protein Gels	Invitrogen	Cat# NP0323BOX
Immobilon-p PVDF membrane	Millipore	Cat# IPVH00010
Pierce BCA protein assay kit	Thermo Fisher Scientific	Cat# 23225
4× Laemmli sample buffer	BioRad	Cat #1610747
CellTiter 96 Aqueous One Solultion Cell Proliferation assay	Promega	Cat# G4100
CellTiter-Glo® 2.0 Cell Viability Assay solution	Promega	Cat# G9243
RNeasy Mini Kit	QIAGEN, Hilden, Germany	74104
RIPA lysis buffer	Millipore-Sigma, St. Louis, MO, USA	R0278
Protease inhibitors	Thermo Scientific, Waltham, MA, USA	A32955
BCA assay	Thermo Scientific, Waltham, MA, USA	23235
SDS-PAGE	Invotogen, Thermo Fisher Scientific, MA, USA	NW04122BOX
SuperSignal™ West Pico PLUS Chemiluminescent Substrate	Thermo Scientific, Waltham, MA, USA	34580
TRIzolTM Reagent	Invitrogen, Waltham, MA, USA	15596026
Phosphate-buffered saline	Thermo Scientific, Waltham, MA, USA	14190250
Trichrome Stain Kit	Abcam Limited, Cambridge, UK	ab150686

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
RNA-seq data generated in this work	This paper	GEO: GSE291867, GSE291190, and GSE291833.
Code and data associated with the computational pipeline	This paper	Zenodo repository: https://doi.org/10.5281/zenodo.15198638
Experimental models: Cell lines		
Human: Huh7	Mark Kay (Stanford University)	Cellosaurus RRID: CVCL_0336
Human: HepG2	ATCC	Cat# HB-8065, Cellosaurus RRID: CVCL_0027
Human: Hep3B	ATCC	Cat# HB-8064, Cellosaurus RRID: CVCL_0326
Human: Primary hepatocytes	Sekisui XenoTech	Cat# CHP48
Experimental models: Organisms/strains		
Mouse: NOD.Cg-Prkdcscid Il2rgtm1Wjl/SzJ	The Jackson Laboratory	Strain #:00555, RRID:IMSR_JAX: 005557
Mouse: C57BL/6J	The Jackson Laboratory	Strain #:000664 RRID:IMSR_JAX: 000664
Oligonucleotides		
UHRF1 siRNA	Thermo Fisher Scientific	Cat# s26553
UHRF1 siRNA	Thermo Fisher Scientific	Cat# s26554
UHRF1 siRNA	Thermo Fisher Scientific	Cat# s26555
Recombinant DNA		
Software and algorithms		
Graphpad Prism v. 10	https://www.graphpad.com/scientific-software/prism/	RRID:SCR_002798
Image Studio lite 5.2.5	https://www.licor.com/bio/image-studio/	RRID:SCR_013715
GPS source code	This paper	https://github.com/Bin-Chen-Lab/GPS
GPS web portal	This paper	http://apps.octad.org/GPS/
OCTAD web portal	Zeng et al.[27]	http://octad.org/
OCTAD R package	Bioconductor	https://bioconductor.org/packages/release/bioc/html/octad.html
Other		
Countess™ Cell Counting Chamber Slides	Thermofisher	Cat# C10283